5

Introductory Quantum Mechanics for Nanoscience

5.1 INTRODUCTION

There are often advantages of making things smaller. Especially, in modern semiconductor industry, miniaturization of devices often delivers more efficient performance. But the million dollar question is what would be the limits of miniaturization? Theoretically, devices can be miniaturized to atomic level, whose sizes can be varied from fraction of a nanometre to few hundreds of nanometres. 'Nanotechnology' deals with these kinds of human-designed working devices produced in a controlled and reproducible manner. The idea of miniaturization of the size-scale is fundamentally interesting due to the fact that as size approaches the atomic scale, the relevant physics associated with the systems changes from classical domain to quantum-mechanical regime. These changes in the behaviours from classical to atomic scale via 'mesoscopic' domain can be explained by modern quantum-mechanical treatments. But in several specific cases, more complex treatments and new methodologies are needed to explain the behaviour of the system. It should be noted that these changes from classical physics to quantum-mechanical nanophysics may mean that some existing devices will fail, while the same changes can open up possibilities for new devices.

In fact, both the natural as well as the artificial world can now be categorized in two regimes: micro-regime and nano-regime. Starting from a human hair to DNA structure—the nature evolves itself from micro- to nano-scale structures. Similarly, man-made world is now shifting its attention from micro devices to nanomaterials. Figure 5.1 schematically represents the broad spectrum of the micro- and nano-regime, indicating how natural and man-made world evolve into smaller domain.

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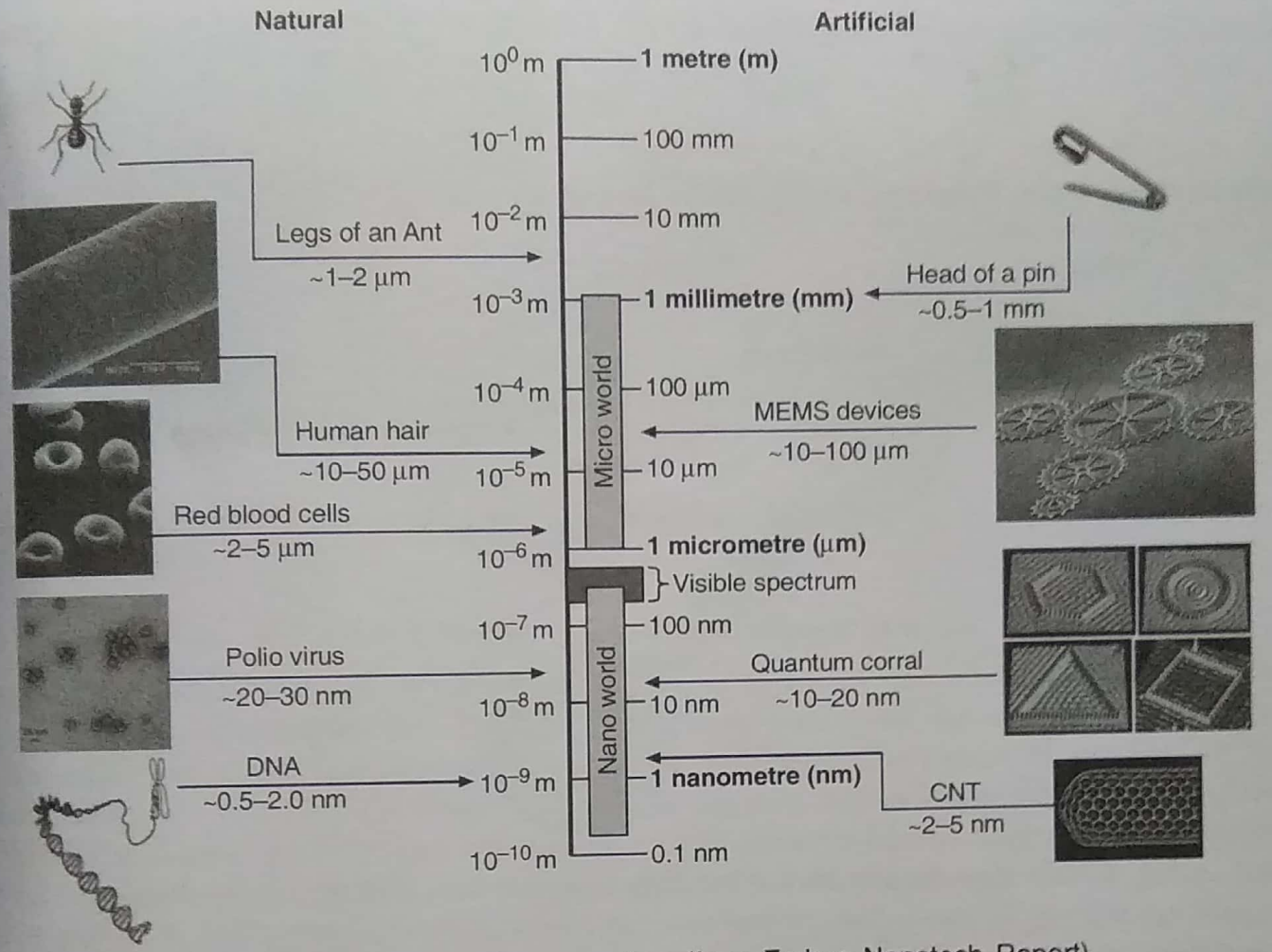


Fig. 5.1 Description of nano-regime (from Forbes Nanotech Report).

5.2 SIZE-EFFECTS IN SMALLER SYSTEMS: PRE-QUANTUM

1. Response time increases for miniaturized simple pendulum

Many artificial systems show interesting behaviour when these systems are scaled down to smaller sizes. For example, the fundamental frequency of a simple harmonic oscillator increases for smaller systems. The resonance frequency of a simple pendulum is given by

$$\nu = \frac{1}{2\pi}\sqrt{\frac{g}{l}} \tag{5.1}$$

where g is the gravitational acceleration and l is the length of the pendulum. This shows that a simple pendulum of length 1 μm will generate 1000 Hz tone, corresponding to a time period of 1 ms. Therefore the miniature version of the simple pendulum has a much faster response time with respect to a normal grandfather pendulum which would have a resolution of 1 s for a pendulum length of 1 m.

2. Thermal time constant decreases in smaller systems

If a hot body (at $T > 0$ K) of heat capacity per unit volume C, is connected to a cold body (at $T = 0$ K) through a heat channel of length l, cross-section a, and thermal heat capacity k_{Th}

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(Fig. 5.2a), then the rate of flow of heat $d\Theta/dT$, from hot to cold body can be written as

$$\frac{d\Theta}{dT} = \frac{k_{Th} aT}{l} \tag{5.2}$$

and at steady state, this equals to the rate of loss of heat from the hot body.

$$\frac{d\Theta}{dT} = -CV\frac{dT}{dt} \tag{5.3}$$

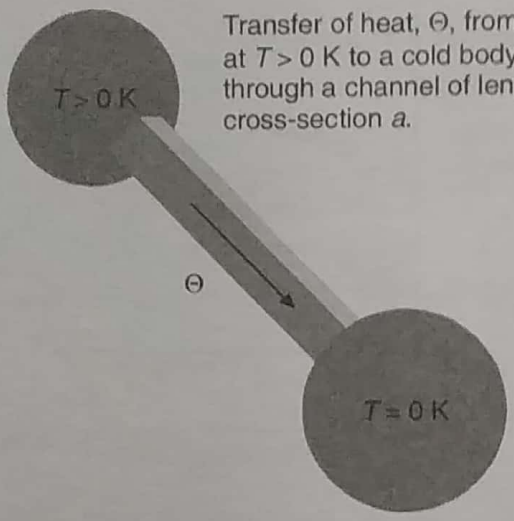


Fig. 5.2(a) Heat transfer from a hot body to a cold body through a heat-channel.

where, V is the volume of the hot body and negative sign indicates the loss of heat from the hot body. Hence, from Eqs. (5.2) and (5.3), we get

$$-CV\frac{dT}{dt} = \frac{k_{Th} aT}{l} \tag{5.4}$$

and

$$\frac{dT}{T} = -\left(\frac{k_{Th} a}{lCV}\right)dt \tag{5.5}$$

which leads to the solution

$$T = T_0\, e^{-t/\tau_{th}} \tag{5.6}$$

where, $\tau_{th} = lCV/k_{Th}a$ is the thermal time constant, which under isotropic environment has the dimension of $[l^2]$. Therefore, the thermal time constant strongly decreases with the size of the system. This phenomenon is used for data storage system where the data bits are recorded with a resistively heated AFM tip and are detected by the integrated piezoelectric PZT sensor. This system consumes less power and responds much faster [1]. Figure 5.2(b) shows the image of a AFM tip and its working principle [2].

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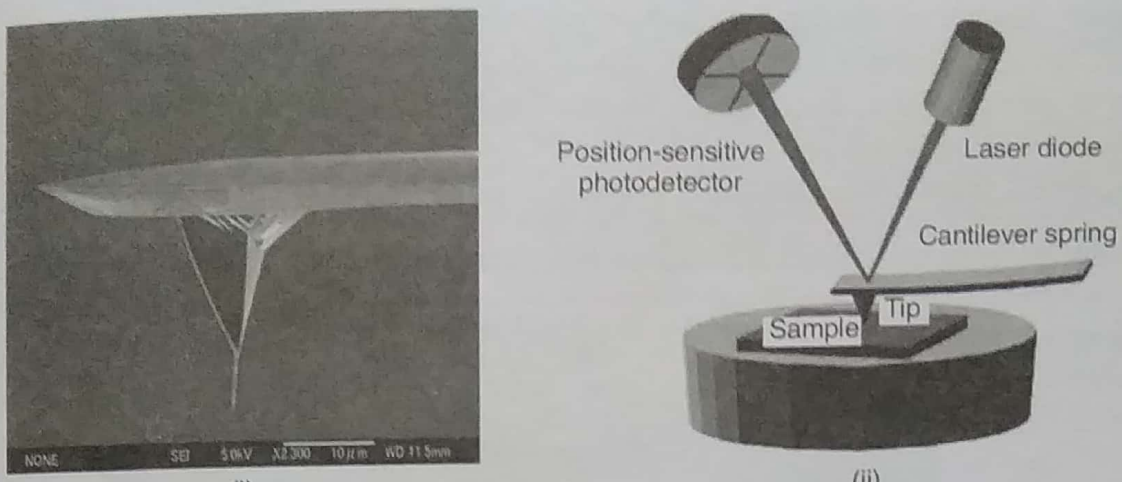


Fig. 5.2(b) (i) SEM image of AFM tip, and (ii) schematic diagram of the working principle of AFM.

3. Disappearance of friction in highly symmetric molecular systems

In molecular-scale systems, the viscous and frictional forces are assumed to be zero, as envisioned by Eric Drexler [3]. Theoretical modelling shows that the self-alignment of the nanometer-scale wheels and axles in vacuum leads to absolutely zero space for any fluid in the interface between two surfaces. Therefore these moving parts will no longer experience any fluid viscous forces between them, but may encounter some frictional forces (if any) at a much lower level than ordinary mechanical frictional forces observed in macroscopic moving parts. For example, in nested carbon nanotubes, which are essentially rolled sheets of graphite (as shown in Fig. 5.3), with no dangling bonds perpendicular to their surfaces, there are no molecules between the surfaces of two adjacent tubes, and, thus the medium between the very closely spaced moving elements is vacuum. So the motion between two neighbouring tubes is almost frictionless.

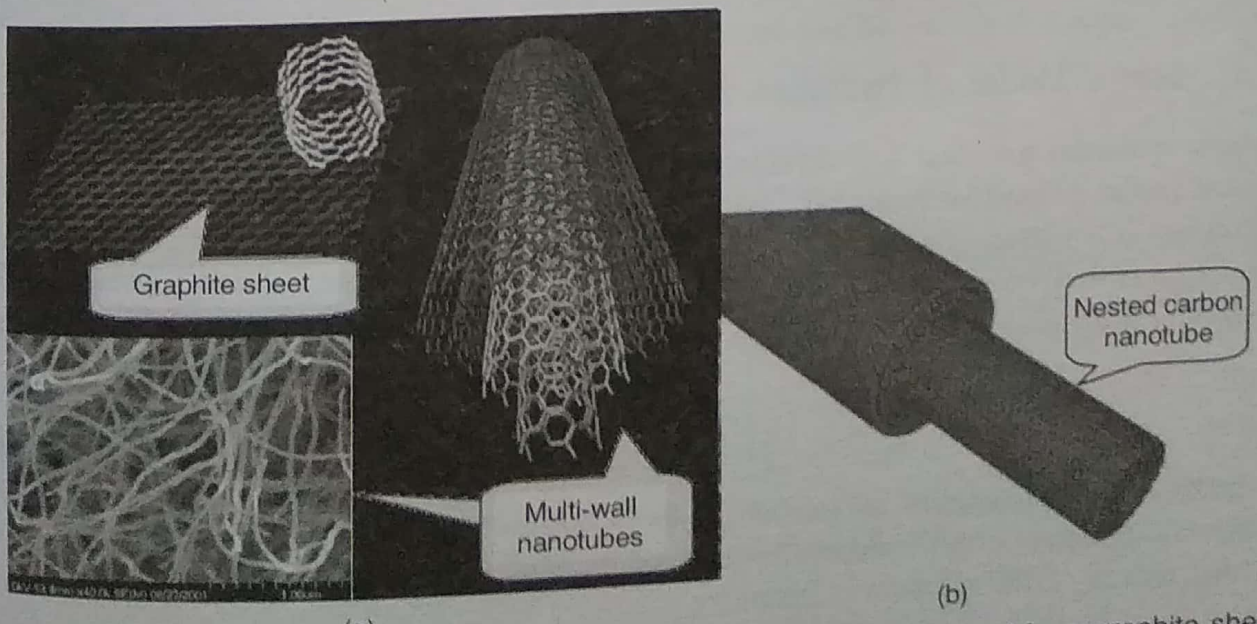


Fig. 5.3 Schematic representation of **(a)** multi-wall carbon nanotubes formed from graphite sheets, and **(b)** computer generated model of nested carbon nanotube.

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Another example of friction-free motion can be found in nature, which is the molecular bearing in biological rotary motors [4], which rotate flagella to move cells in liquid. The flagellum is attached to a shaft, which rotates freely in a molecular bearing structure. The rotation transmits torque and power and the motors operate continuously over the life of the cell, which suggests a friction-free molecular bearing. A schematic representation of E.coli bacterial flagella rotary motor is given in Fig. 5.4.

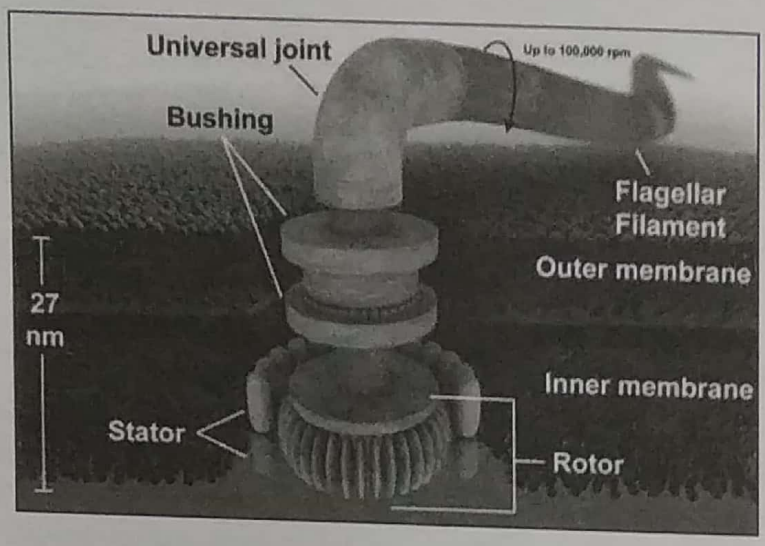


Fig. 5.4 Schematic representation of *E.coli* bacterial flagella rotary motor.

5.3 QUANTUM BEHAVIOUR OF NANOMETRIC WORLD

In atomic scale, the failure of classical concepts leads to the quantum-mechanical approach to understand the behaviour of matters in such low-dimensions. A clear understanding of these nano-physical rules is the key to understand the deviations of the devices and machines that will appear as their sizes are reduced towards atomic scale.

5.3.1 Bohr's Model of Hydrogen Atom

The first quantum rule that has diverse impact on the quantum physics is the Bohr's semi-classical model of nuclear atom [5]. This model is semi-classical in the sense that the motion of electrons is assumed to be planetary in nature, and, thus the energy of an electron, encircling the nucleus is derived according to the classical physics. Therefore, a single electron of charge $-e$, circling a massive nucleus of charge $+Ze$ is subjected to a coulombic attraction of

$$F = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2} \tag{5.7}$$

where ε_0 is the permittivity of free space ($= 9 \times 10^9\ \mathrm{Nm^2C^{-2}}$). This force is balanced by centripetal acceleration (v^2/r) times the mass of the electron ($m_e = 9.1 \times 10^{-31}$ kg). Therefore, the total energy of the electron is given by

$$E = \frac{m_e v^2}{2} - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2r} \tag{5.8}$$

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where, r is the radius of electron orbit and v is the velocity of the electron at a certain position and time.

But this classical relation between the energy of an electron in an orbit of radius r, predicts a collapse of the motion as for small r the energy is increasingly favourable (more negative), which indicates that the electron will eventually spiral into the nucleus giving off its energy as electromagnetic radiation. Thus all the atoms and, in fact, all matters are unstable on the basis of this classical prediction.

As this type of collapse does not happen in nature, Bohr's model assumed the quantization of angular momentum (L) of the electron of mass m_e, circling the nucleus in an orbit of radius r, with a speed v, according to the relation

$$L = m_e v r = \frac{nh}{2\pi} \tag{5.9}$$

where h is the Planck's constant and n is the arbitrary quantum number ($n = 1, 2, 3,\ldots$, etc). Thus electrons are stabilized in the 'Bohr orbits' of hydrogen and similar kind of one-electron atoms. The energy and radius of the nth orbit can be written as

$$E_n = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2r_n} \tag{5.10}$$

and

$$r_n = \frac{n^2 a_0}{Z} \tag{5.11}$$

where $a_0 = h^2\varepsilon_0/\pi m_e e^2 = 0.053$ nm is the Bohr radius of hydrogen atom and the energy of the electron in the first Bohr orbit ($n = 1$) of the hydrogen atom ($Z = 1$) is given by

$$E_1 = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{2r_1} = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{2a_0} = -\frac{1}{8\varepsilon_0^2}\frac{e^4 m_e}{h^2} = -13.6\ \text{eV} \tag{5.12}$$

Hence the energy of nth orbit of similar one-electron atom can be written in terms of first Bohr orbit as

$$E_n = E_1\left(\frac{Z^2}{n^2}\right) \tag{5.13}$$

This equation is nicely used to determine the discrete emission and absorption lines in one-electron atomic spectra by the following quantum condition

$$\Delta E = E_f - E_i = \frac{hc}{\lambda} = E_1 Z^2\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \tag{5.14}$$

where the energy change of the one-electron atom goes from an initial quantum number n_i to a final quantum number n_f. Figure 5.5 shows the spectral lines originated in transitions between energy levels of a hydrogen atom.

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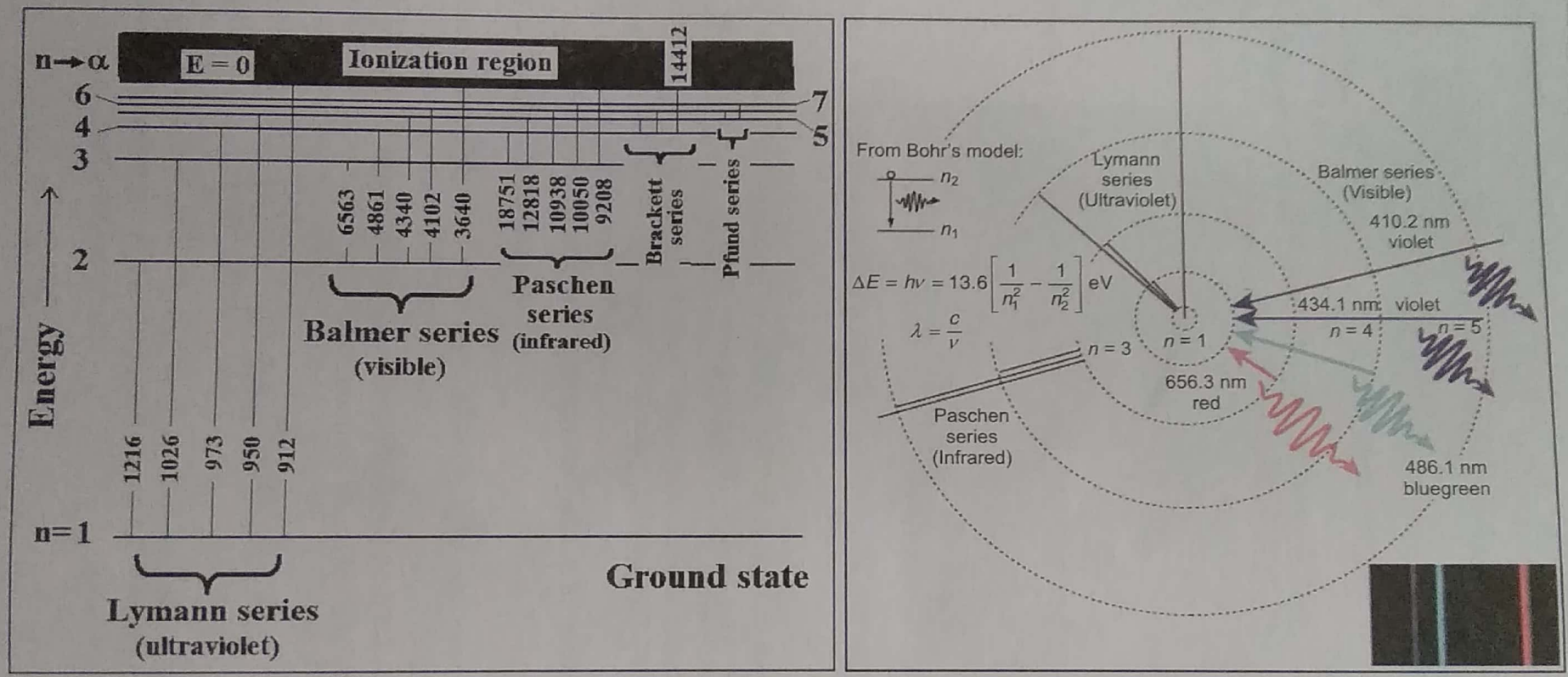


Fig. 5.5 **(a)** Hydrogen line spectra, and **(b)** description of the transitions various Bohr orbits.

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In nanophysics, the formation of 'excitons' can be described by the mathematics of Bohr model (but in an elementary way. When the excitons are generated inside a semiconducting material, more mathematics are involved, which is described latter). An 'exciton' is a bound state of a Coulomb-correlated electron–hole pair in an insulator or semiconductor, generated by irradiation. The electron and hole in this electron–hole in the pair momentarily orbit around each other and emit a photon, which can be described by the relevent Bohr's model.

Also the size-effects on the light emission of the 'quantum-dots' (QD) can be described on the basis of Bohr model. QD is a quantized electronic structure in which electrons are confined with respect to motion in all three dimensions. By changing the size of the QDs, one can generate different wavelengths of light through them under irradiation. Figure 5.6 shows the CdSe QDs of different sizes, emitting different colors.

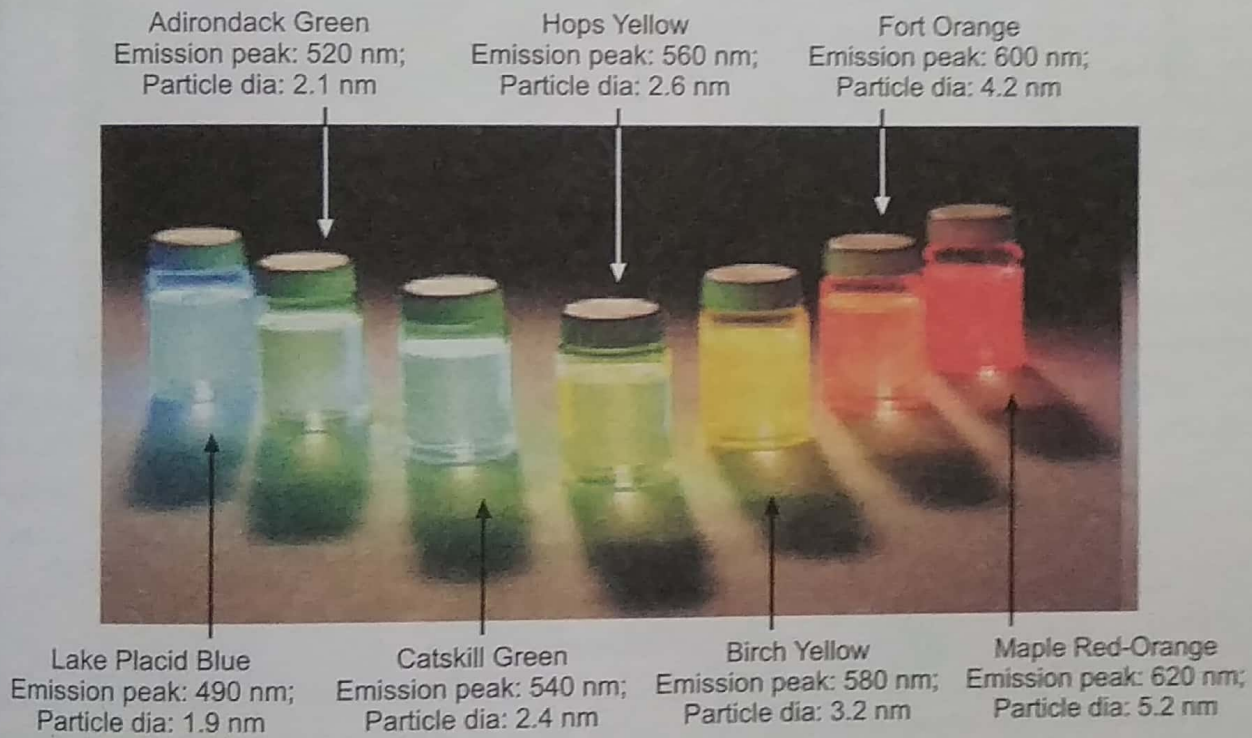


Fig. 5.6 Images of core-shell CdSe/ZnS quantum dots stored in vials, showing emission of different colours with change in the size of the QDs. Data collected from evident technologies [6].

5.3.2 Wave-particle Duality: de Broglie Wavelength

The wave nature of light was established by Young's double slit experiment where the sinusoidal interference pattern of coherent light falling on a screen behind two linear slits of small spacing d, is guided by

$$n\lambda = d \sin \theta \tag{5.15}$$

where λ is the wavelength of light and θ is the angular position of the appearance of maxima on the interference pattern and n is an integral number. The path difference of the light from two slits shall be an integral multiple of light wavelength $n\lambda$, and governs the constructive or destructive patterns on the screen, as shown in Fig. 5.7.

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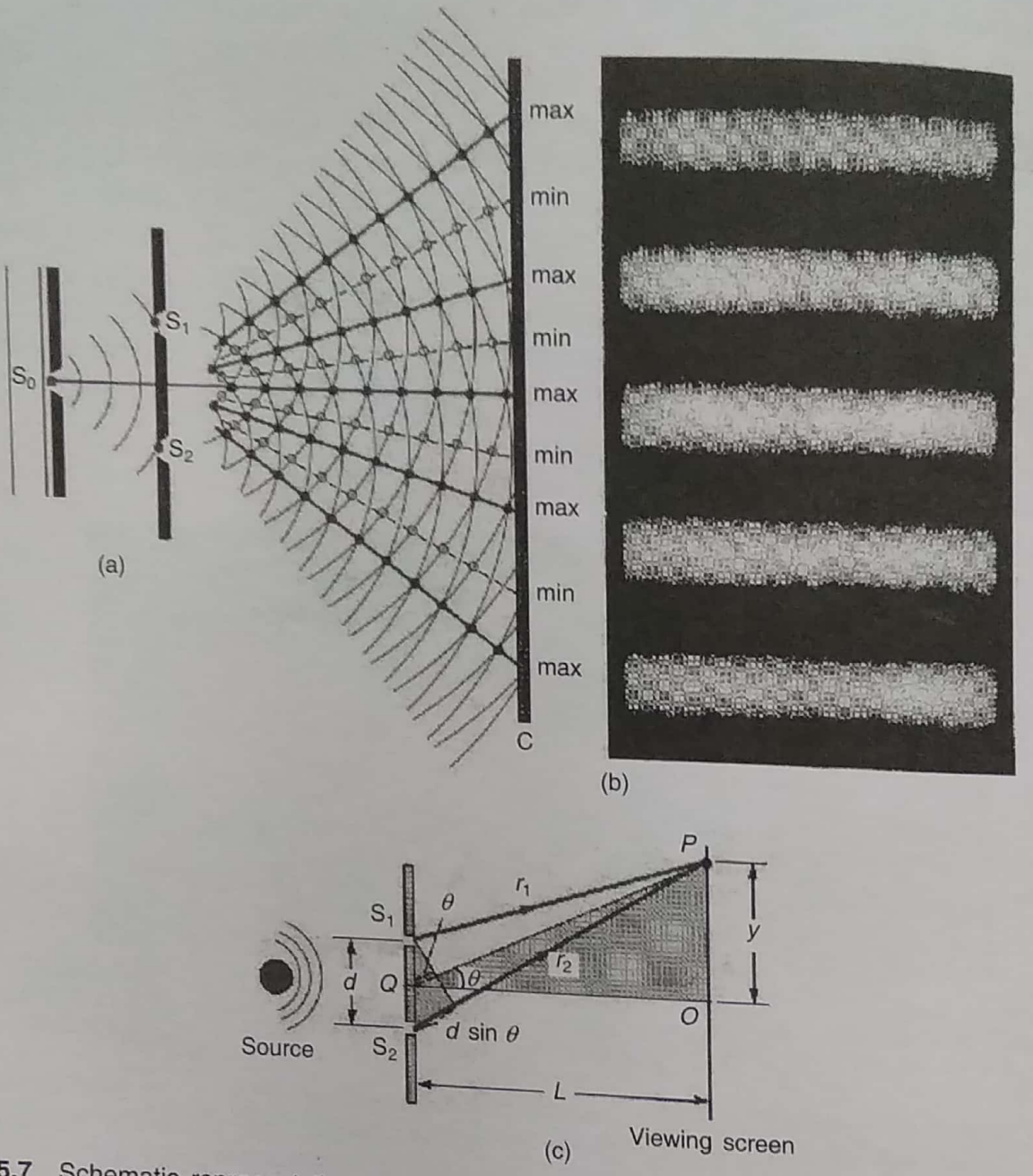


Fig. 5.7 Schematic representation of Young's double slit experiment showing interference pattern.

On the other hand, explanation of the spectrum of 'black body radiation' by Max Planck in 1900, that vibrating atoms that emit light do so in discrete amounts, first postulated that the energy to be quantized, given by the relation

$$E = nh\nu \tag{5.16}$$

where n is an integer, h is the Planck's constant ($= 6.62618 \times 10^{-34}$ J-s).

Later, Einstein in 1905, boldly proposed that light actually consisted of and propagated in discrete energy packets called photons, with the energy of each photon being related to the wave frequency by Planck's relation (Eq. (5.16)). Indeed it is for his explanation of the photoelectric effect (Fig. 5.8) using the photon concept that Einstein was awarded the Nobel Prize in 1921. Note how the discrete particle nature of photons nevertheless depends on the frequency of a wave as shown in Eq. (5.16).

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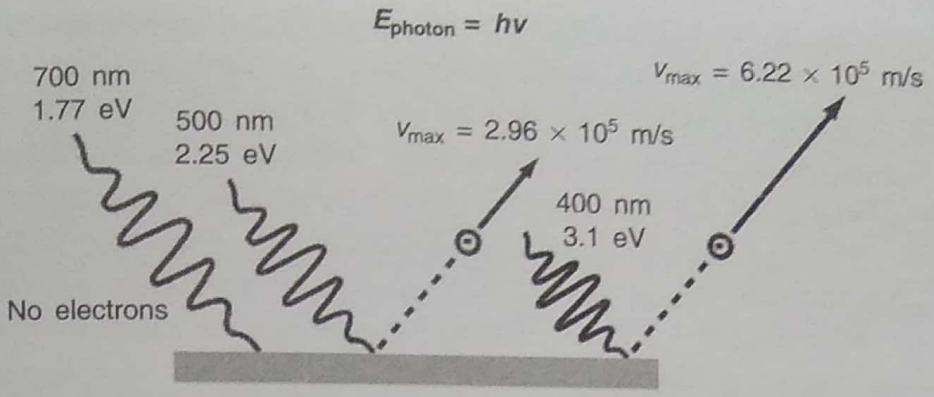
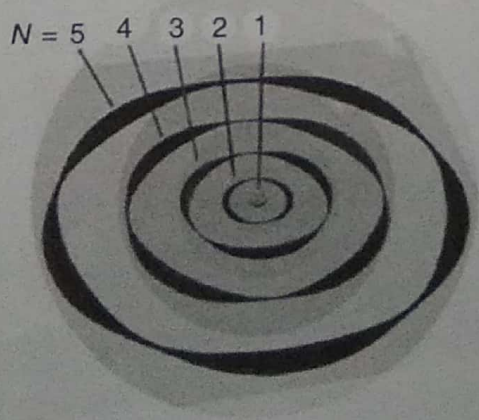


Fig. 5.8 Explanation of particle nature of light in photoelectric effect. The following algebraic relation $hv = \phi + KE_{max}$ explains that a part of the incident photon energy is used to overcome the work function ($\phi = hv_0$, v_0 = threshold frequency) of the material to extract a photoelectron and rest is converted to the kinetic energy of the photoelectron ($1/2\ mv^2_{max}$). Therefore, one can rewrite the above relation as $hv = hv_0 + (1/2)\ mv^2_{max} \Rightarrow h(v - v_0) = (1/2)\ mv^2_{max}$. Therefore, below the threshold frequency there will be no photo-electron emission, however, intense the incident light might be.

Since light has both wave-like and particle-like nature, Louis de Broglie postulated the similar wave-like nature of matters. In physics, the de Broglie hypothesis is the statement that all matter (any object) has a wave-like nature (wave-particle duality). The de Broglie relations show that the wavelength is inversely proportional to the momentum of a particle and that the frequency is directly proportional to the particle's kinetic energy:

$$\lambda = \frac{h}{p} = \frac{h}{\gamma m_0 v} = \frac{h}{m_0 v}\sqrt{1 - \frac{v^2}{c^2}} \tag{5.17}$$

where h is the Planck's constant, m_0 is the particle rest mass, v is the particle velocity, γ is the Lorentz factor, and c is the speed of light in vacuum. The hypothesis was advanced by Louis de Broglie in 1924 in his Ph.D. thesis [7]; he was awarded the Nobel Prize for physics in 1929 for this work, which made him the first person to receive a Nobel Prize on a Ph.D. thesis. Figure 5.9 shows the wave nature of electrons in the Bohr orbits of an atom.

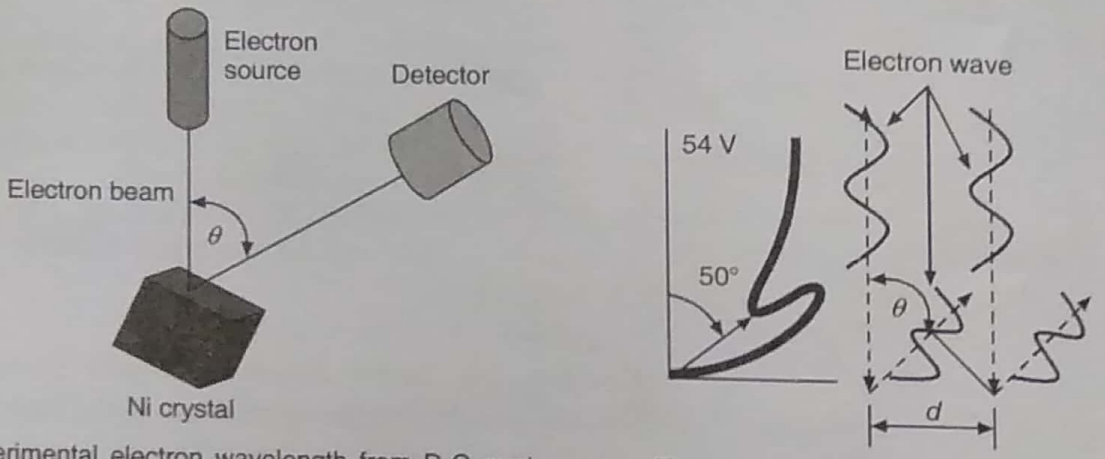


Bohr–de Broglie electron matter-waves. Electrons have wave-like nature as shown for shells 1 to 5

Fig. 5.9 Schematic representation of wave-like nature of the electron in Bohr orbits.

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This wave-particle duality of matters was confirmed by Davisson and Germer by their famous electron-diffraction experiment [8]. This experiment demonstrated the wave nature of the electron, confirming the hypothesis of de Broglie. Putting wave-particle duality on a firm experimental footing, it represented a major step forward in the development of quantum mechanics. The Bragg's law for diffraction had been applied to X-ray diffraction, but this was the first application to particle waves [9]. The details of the observed diffraction pattern could be well-fitted with the theoretical value if the wavelength associated with the electrons were exactly given by (h/p). Figure 5.10 schematically represents the Davisson–Germer experiment showing the close matching of experimentally obtained electron wavelength with theoretical



Experimental electron wavelength from D-G expt.:
Path length difference: $d \sin\theta = 2.15 \sin 50° = \lambda$
$= 1.65$ Å for constructive interference.
(Ni lattice spacing, $d = 2.15$ Å)

Theoretical electron wavelength from de Broglie hypothesis:
$\lambda = h/mv = 1.67$ Å for 54 V

Fig. 5.10 Schematic representation of Davission–Germer experiment. A 54 V electron wavelength nearly matches with experimentally obtained electron wavelength diffracted through Ni lattice.

value. On the other hand, Figs. 5.11(a) and (b) show the diffraction pattern of a beam of electrons through a thin Al foil, indicating wave-nature of matters. Next came the concept of wave function ψ, to represent wave properties of the matters by some equations in a given situation.

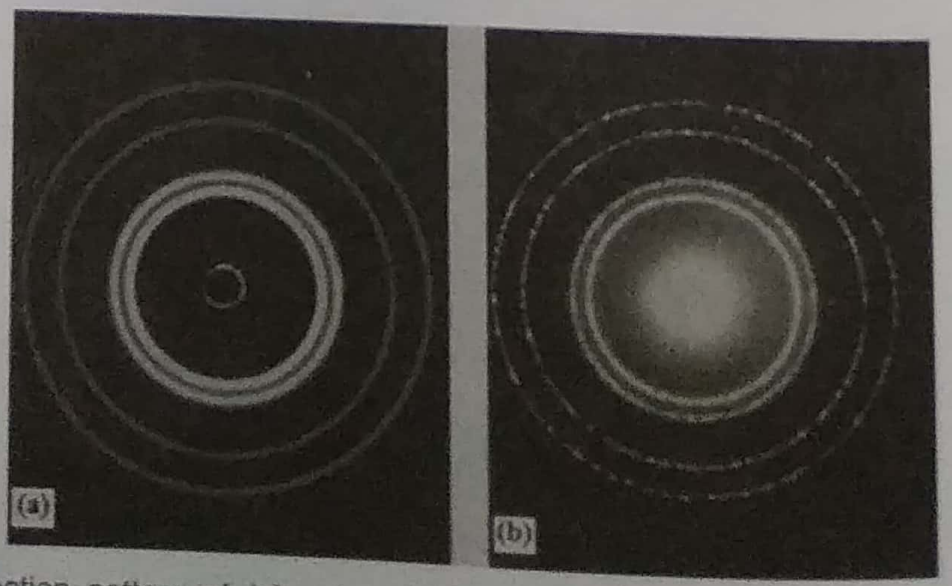


Fig. 5.11 Diffraction pattern of **(a)** a beam of X-ray passing through thin aluminium foil, and **(b)** by a beam of electrons passing through the same foil.

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5.3.3 Wave Function Associated with an Electron

A microscopic particle is guided by a wave function ψ, which is assumed to be a function of position vector $\vec{r}$, and time t, such that

$$\psi = \psi(\vec{r}, t) \tag{5.18}$$

In general, ψ is a complex number and the probability of finding the particle is given by

$$\psi^*(\vec{r},t)\,\psi(\vec{r},t) = P(\vec{r},t) \tag{5.19}$$

where $P(\vec{r},t)$ is called the probability density. Therefore, the net probability of finding the particle in a particular volume, $dV = dx\,dy\,dz$ is given by

$$P(\vec{r},t)\,dx\,dy\,dz = \psi^*(\vec{r},t)\,\psi(\vec{r},t)\,dx\,dy\,dz \tag{5.20}$$

The wave function should be chosen in such a way that $P(\vec{r},t)$ should be normalized. That is, Eq. (5.20) should be integrated over whole space such that the particle should exist in that region. Therefore

$$\iiint_{x,y,z} P(\vec{r},t)\,dx\,dy\,dz = \iiint_{x,y,z} \psi^*(\vec{r},t)\,\psi(\vec{r},t)\,dx\,dy\,dz = 1 \tag{5.21}$$

This is also called the normalization of the wave function.

For a beam of particles of identical energy ($= p^2/2m$) moving in one-dimension (say, along X-axis) is considered to be a travelling wave as

$$\psi(x,t) = \sqrt{\frac{1}{L}} \times e^{i(kx-\omega t)} = \sqrt{\frac{1}{L}}\,[\cos(kx-\omega t) + i\sin(kx-\omega t)] \tag{5.22}$$

where $k = 2\pi/\lambda$ is the angular wave-number and $\omega = 2\pi\nu$ is the angular frequency and $\sqrt{1/L}$ is the normalization factor which physically means the presence of one particle in each length L, along an infinite x-axis.

Also, according to the de Broglie relation

$$p = \frac{h}{\lambda} = \frac{hk}{2\pi} = \hbar k \tag{5.23}$$

and

$$E = h\nu = \frac{h\omega}{2\pi} = \hbar\omega \tag{5.24}$$

where $\hbar$ is the reduced Plank's constant ($= 1.0546 \times 10^{-34}$ J-s). Therefore, Eq. (5.22) can be rewritten in terms of energy of the particle (E) and momentum (p) as

$$\psi(x,t) = \sqrt{\frac{1}{L}} e^{(i/\hbar)(px-Et)} = \sqrt{\frac{1}{L}}\left[\cos\frac{1}{\hbar}(px-Et) + i\sin\frac{1}{\hbar}(px-Et)\right] \tag{5.25}$$

This equation is more logical for particle wave as it contains physical parameters like energy and momentum, related to matter, rather than wave number and frequency.

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Now it is to be noted that in Eqs. (5.22) and (5.25), a point of fixed phase (such as a peak in the real part of the wave, shown as point 'P' in Fig. 5.12) moves as $x = (\omega/k)t$, so that

$$v_{ph} = \frac{\omega}{k} \tag{5.26}$$

is called the phase velocity. Also $\psi(x, t)$ has a constant absolute value at any x, describing a particle equally likely to be at any position on the infinite X-axis and there is no localization in this wave-function since the momentum is perfectly described, implying $\Delta x \to \infty$!!

Also, from relativistic formula, $E = mc^2$, we can write the phase velocity as

$$v_{ph} = \frac{\omega}{k} = \left(\frac{E}{h}\right)\left(\frac{\lambda}{2\pi}\right) = \frac{(mc^2)}{h}\left(\frac{h}{p}\right) = \left(\frac{mc^2}{mv}\right) = \frac{c^2}{v} \tag{5.27}$$

As particle velocity, $v < c$, the velocity of light, therefore $v_{ph} > c$. That is, the de Broglie waves travel faster than light !!!! To understand this unexpected result, the Heisenberg uncertainty principle comes into play to give the wave description of the location of a particle and introduce the term 'group velocity v_g', to explain the matter wave.

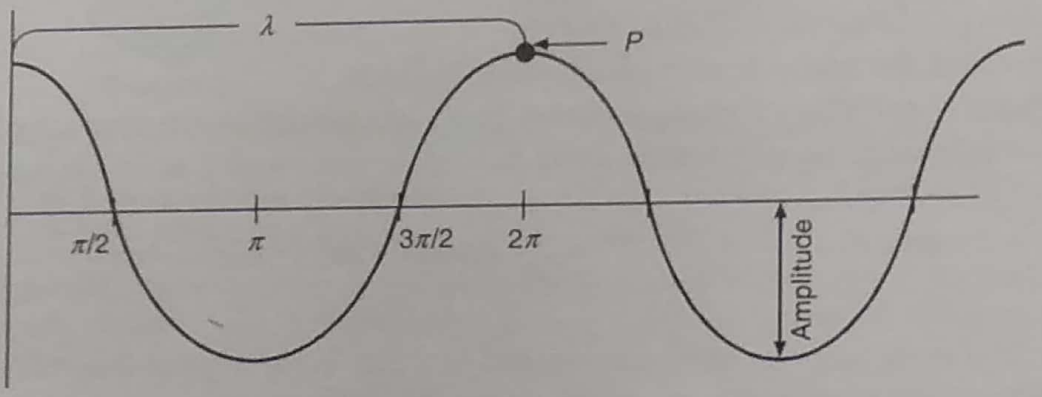


Fig. 5.12 Representation of a cosine wave.

5.3.4 Heisenberg Uncertainty Principle; Description of Matter Wave: Wave Packet

Heisenberg uncertainty principle states that the position x, and momentum p, of a particle can be measured simultaneously only to minimum levels of uncertainty, Δx and Δp respectively, such that

$$\Delta x \Delta p \geq \frac{h}{2} \tag{5.28}$$

The uncertainty principle is basically a consequence of the wave description of the location of a particle. For the wave-function $\psi(x, t)$ described in Eq. (5.22), as the free particle momentum $p = hk$ can be measured precisely, therefore $\Delta p = 0$, which represents a case $\Delta x \to \infty$.

Description of localized particle can be done by a wavefunction which will be a linear combination of waves having a distribution of k and ω values. For example, a linear combination of two waves can be written as

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$$\left.\begin{aligned}\psi(x,t) &= \psi_1(x,t)+\psi_2(x,t)\\ &= A\{\cos(kx-\omega t)+\cos[(k+\Delta k)x-(\omega+\Delta\omega)t]\}\\ &= 2A\cos\left(\frac{\Delta k}{2}x-\frac{\Delta\omega}{2}t\right)\cos(\bar{k}x-\bar{\omega}t)\end{aligned}\right\} \quad (5.29)$$

where $\bar{k} = k + \Delta k/2$, and $\bar{\omega} = \omega + \Delta\omega/2$. Now for $\Delta k \to 0$ and $\Delta\omega \to 0$, the factor $\cos[(\Delta k/2)x - (\Delta\omega/2)t]$ acts like an envelope function of long wavelength and low frequency which modulates the original wave as shown in Fig. 5.13.

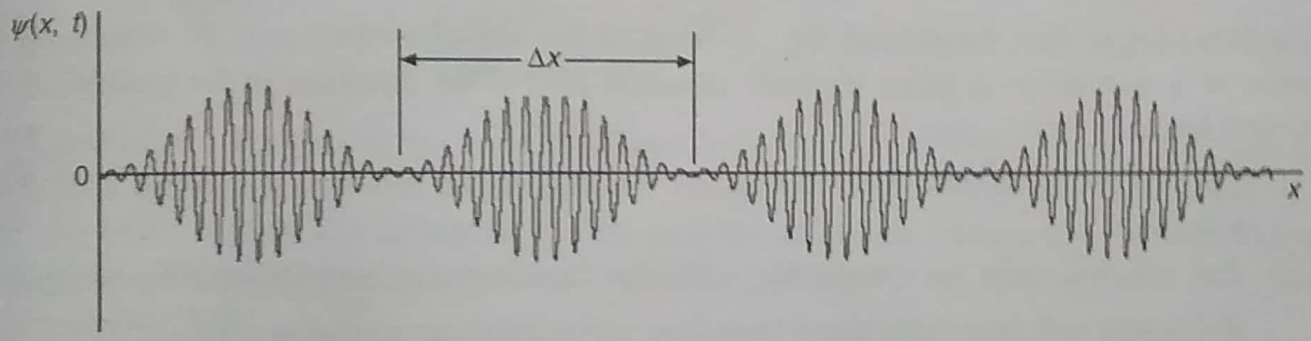


Fig. 5.13 Modulated waves showing regions of localization that move with group velocity.

At a fixed time, t, the distance between two adjacent nodes, Δx, can be considered as the localization length of the particle represented by the superposition of two waves and represented by

$$\left.\begin{aligned}\Delta x &= \frac{2\pi}{\Delta k}\\ \Rightarrow\quad \Delta x\,\Delta k &= 2\pi\end{aligned}\right\} \quad (5.30)$$

This is a form of uncertainty principle, which is basically a wave property. Now according to de Broglie relation, $\Delta p = \hbar\,\Delta k$. This gives

$$\Delta p\,\Delta x = h \quad (5.31)$$

This resembles with Eq. (5.28), but an accurate localization through Gaussian distribution of k values gives a narrower uncertainty value as stated in Eq. (5.28).

The group velocity (v_g) of the wave packet is described by the motion of the envelope function of Eq. (5.29) as

$$v_g = \frac{x}{t} = \frac{\Delta\omega}{\Delta k} \approx \frac{\partial\omega}{\partial k} \text{ at } \Delta k \to 0 \quad (5.32)$$

Now, for a localized moving particle of energy E, we have

$$\left.\begin{aligned}E = \hbar\omega &= \frac{p^2}{2m} = \frac{\hbar^2k^2}{2m}\\ \Rightarrow\quad \omega &= \frac{\hbar k^2}{2m}\end{aligned}\right\} \quad (5.33)$$

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Therefore, the group velocity v_g, will be given by

$$v_g = \frac{\partial \omega}{\partial k} = \frac{\hbar k}{m} = \frac{p}{m} = v \tag{5.34}$$

Thus the group velocity and the de Broglie relation reproduces the classical particle velocity v. Hence, this expression of group velocity has physical significance for particle wave, unlike phase velocity.

5.3.5 Schrödinger Equation

Erwin Schrödinger first formulated the equation of the wavefunction $\psi(x, t)$, describing the location of a particle in a given physical situation [10]. This equation is formulated on the following basis:

(i) The correct wave equation must reflect the de Broglie relation, $\lambda = h/p$ (i.e. $E = \hbar\omega$, $p = \hbar k$).

(ii) The solution must be compatible with the Davisson–Germer experiment of electron diffraction and must represent a travelling wave solution, $\psi(x, t) = \sqrt{1/L}\, e^{(i/\hbar)(px-Et)}$ for a free particle.

To formulate the correct wave equation, the electromagnetic wave equation derived from Maxwell's equation has been taken as the guiding direction. This wave equation is a second order partial differential equation of electric field $\vec{E}$ and magnetic field $\vec{H}$ as

$$\left.\begin{aligned}\left(\nabla^2 - \mu_0\varepsilon_0 \frac{\partial^2}{\partial t^2}\right)\vec{E}(\vec{r},t) = 0\\ \left(\nabla^2 - \mu_0\varepsilon_0 \frac{\partial^2}{\partial t^2}\right)\vec{H}(\vec{r},t) = 0\end{aligned}\right\} \tag{5.35}$$

where $\mu_0 e_0 = 1/c^2$; c is the speed of light, indicating that the origin of electromagnetic wave as similar to light.

Applying the above equation (in one-dimension) for the travelling wave solution $\psi(x, t) = \sqrt{1/L}\, e^{(i/\hbar)(px-Et)}$, we get

$$\left(\frac{\partial^2}{\partial x^2} - \mu_0\varepsilon_0 \frac{\partial^2}{\partial t^2}\right)\Psi(x,t) = 0$$

or

$$\left(\frac{\partial^2}{\partial x^2} - \mu_0\varepsilon_0 \frac{\partial^2}{\partial t^2}\right)\sqrt{\frac{1}{L}}\, e^{(i/\hbar)(px-Et)} = 0$$

or

$$(-p^2 + E^2\mu_0\varepsilon_0)\,\Psi(x,t) = 0 \tag{5.36}$$

The factors in the bracket relates to the speed of light as $E/p = \sqrt{1/\mu_0\varepsilon_0} \Rightarrow \omega/k = c$. This indicates that we need to form a matter-wave equation which should have a bracketed term as

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that given in Eq. (5.36), which is having a physical significance as that of electromagnetic wave equation. Looking at the bracketed term of Eq. (5.36), a reasonable guess of the starting relation may be the expression of energy as Kinetic Energy ($p^2/2m$) + Potential Energy (U) = Total Energy (E), which can be converted into the following expression, when substituted with de Broglie relation:

$$\frac{\hbar^2 k^2}{2m} + U = \hbar\omega \tag{5.37}$$

Therefore, similar to Eq. (5.36), one can generate an equivalent equation as

$$\left(\frac{\hbar^2 k^2}{2m} + U - \hbar\omega\right)\Psi(x,t) = 0 \tag{5.38}$$

Now, knowing the solution of the wave-function to be $\Psi(x, t) \sim e^{(i/\hbar)(px - Et)} = e^{i(kx - \omega t)}$, one can reproduce a wave-equation which should involve a term $\partial^2/\partial x^2\ \Psi(x, t)$ (for introduction of k^2 term) and $(\partial/\partial t)\ \Psi(x, t)$ (to introduce ω term into the equation). Hence, the ***one-dimensional time-dependent Schrödinger equation*** can be written as

$$-\frac{\hbar^2}{2m} \times \frac{\partial^2}{\partial x^2}\Psi(x,t) + U(x,t)\,\Psi(x,t) = i\hbar\frac{\partial}{\partial t}\Psi(x,t) \tag{5.39}$$

In 3D, this equation will look like

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + U(\vec{r},t) - i\hbar\frac{\partial}{\partial t}\right]\Psi(\vec{r},t) = 0 \tag{5.40}$$

When the potential function is independent of time, i.e. $U = U(x)$ (in 1D), then spatial and time dependent parts of the total wave function $\Psi(x, t)$ can be separated as

$$\Psi(x,t) = \psi(x)\,\phi(t) \tag{5.41}$$

and when substituted in Eq. (5.39), we get the time dependent part as

$$\phi(t) = e^{-(i/\hbar)(Et)} \tag{5.42}$$

and the spatial part as

$$-\frac{\hbar^2}{2m} \times \frac{d^2}{dx^2}\psi(x) + U(x)\,\psi(x) = E\,\psi(x) \tag{5.43}$$

which is the ***time-independent Schrödinger equation*** (in 1D).

The solution, $\psi(x)$ of this equation must satisfy the boundary conditions and physical requirements. These requirements are:

1. $\psi(x)$ must be continuous,
2. $\psi(x)$ should have a continuous derivative (unless the potential is infinite),
3. $\psi(x)$ should be single-valued and smooth as the probability of finding the particle is $|\psi(x)|^2$.

These requirements are schematically described in Fig. 5.14.

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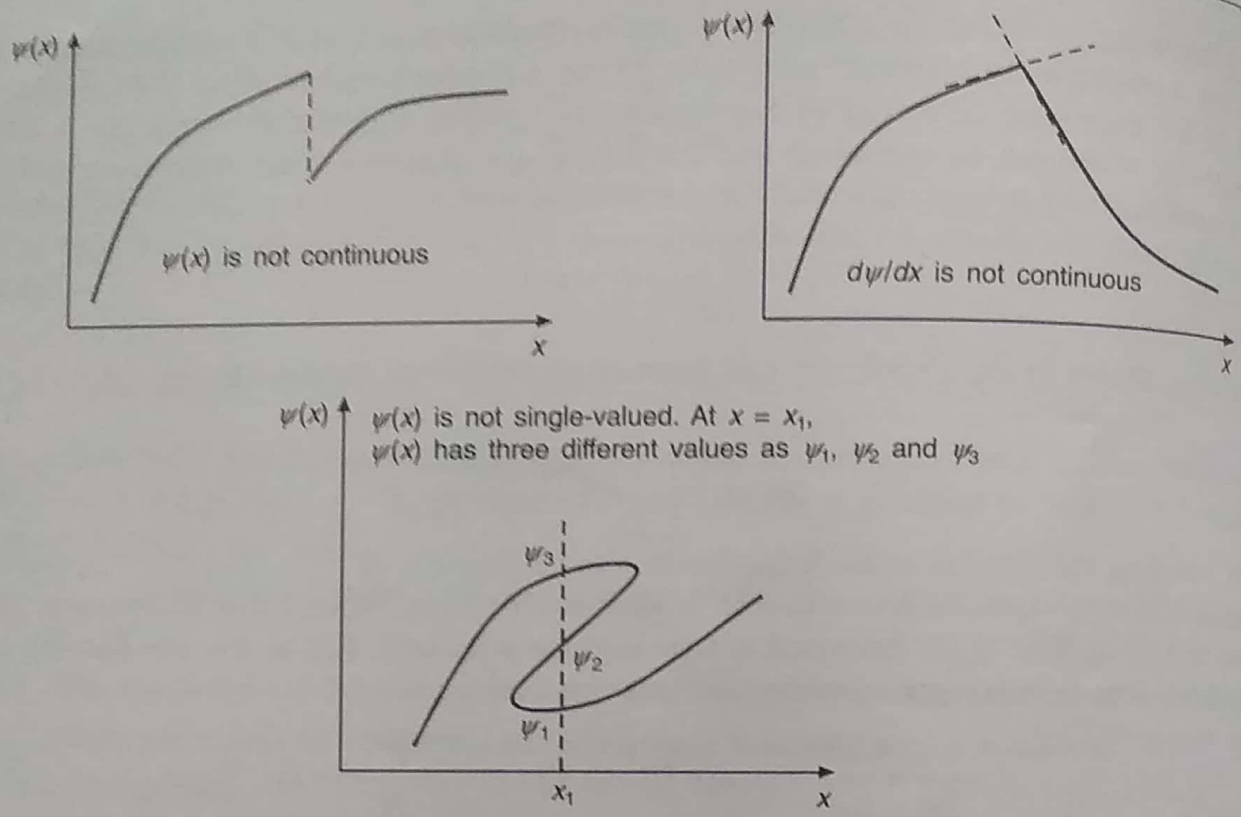


Fig. 5.14 Unacceptable results of $\psi(x)$.

5.3.6 Applications of Schrödinger Equation

(a) Infinite potential well: A confined particle in 1D

A particle is trapped in 1D according to the following potential distribution:

$$U = 0 \text{ for } 0 < x < L$$

$$U \to \infty \text{ for } x \geq L \text{ and } x \leq 0.$$

Schematically, this can be described as in Fig. 5.15.

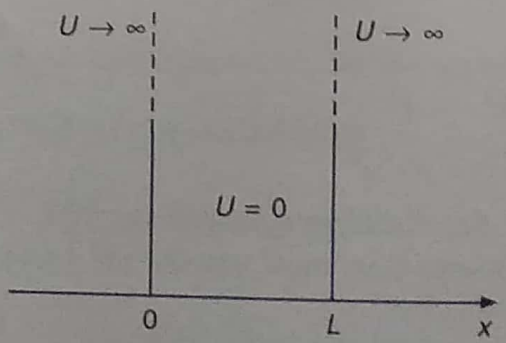


Fig. 5.15 Particle in an infinite quantum well.

This indicates that the probability of finding the particle outside the well is zero. The particle will need infinity energy to come out of the well. Therefore, $\psi(x) = 0$ for $x \leq 0$ and $x \geq L$ and finite inside the well. Hence the Schrödinger equation becomes

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$$\frac{d^2}{dx^2}\psi(x)+\frac{2mE}{\hbar^2}\psi(x)=0 \tag{5.44}$$

The general solution of this second-order linear differential equation will be

$$\psi(x)=Ae^{ikx}+Be^{-ikx} \tag{5.45}$$

where A and B are constants and $k^2 = 2mE/\hbar^2$, where k is a real quantity. Now, considering the boundary conditions $\psi(0) = \psi(L) = 0$ and $d/dx\,(\psi(x)|_{x=0}) = d/dx\,(\psi(x)|_{x=L}) = 0$, we get

$$\psi(x)=2Ai\sin kx \tag{5.46}$$

and

$$kL = n\pi, \quad n = 1, 2, 3, \ldots \tag{5.47}$$

These, in turn, give

$$E_n=\frac{n^2h^2}{8mL^2}, \quad n=1, 2, 3, ... \tag{5.48}$$

and

$$\psi_n(x)=A_1\sin\frac{n\pi x}{L}, \quad n=1, 2, 3, \ldots \tag{5.49}$$

where $A_1 = 2iA$, called the normalization constant and can be determined from the probability of finding the particle inside the quantum well as unity. Thus, we get

$$\int_0^L|\psi(x)|^2=1$$

$$\Rightarrow \quad A_1=\left(\frac{2}{L}\right)^{1/2} \tag{5.50}$$

Thus the wavefunction becomes

$$\psi_n(x)=\left(\frac{2}{L}\right)^{1/2}\sin\frac{n\pi x}{L} \tag{5.51}$$

and called the eigenfunction [11]. All $\psi_n(x)$ for $n = 1, 2, 3,\ldots$, constitutes the eigenfunctions of the system. The energies of the system defined by Eq. (5.48) with $n = 1, 2, 3,\ldots$, are called eigenenergies of the system.

The behaviour of a trapped electron inside an infinite potential well can now be summarized pictorially according to Fig. 5.16. As described in Eq. (5.48) and Eq. (5.51), both the energy and wavefunction are dependent on the quantum number, n. The energy is quantized and can take only some finite values which increase as n^2. Therefore, the lowest energy corresponds to $n = 1$, which is called the ground state energy, the minimum energy the electron can possesses. It is to be noted that the lowest energy of the electron cannot be zero due to the uncertainty principle. It should have a certain momentum, so also kinetic energy in the lowest energy state, even though the potential energy is zero. This example of infinite potential well problem illustrates the typical nanophysical behaviour that there are discrete allowed energy levels and wavefunctions of trapped electrons. The wavefunctions do not precisely locate a particle, instead they only provide the statements on the probability of finding a particle in a given range.

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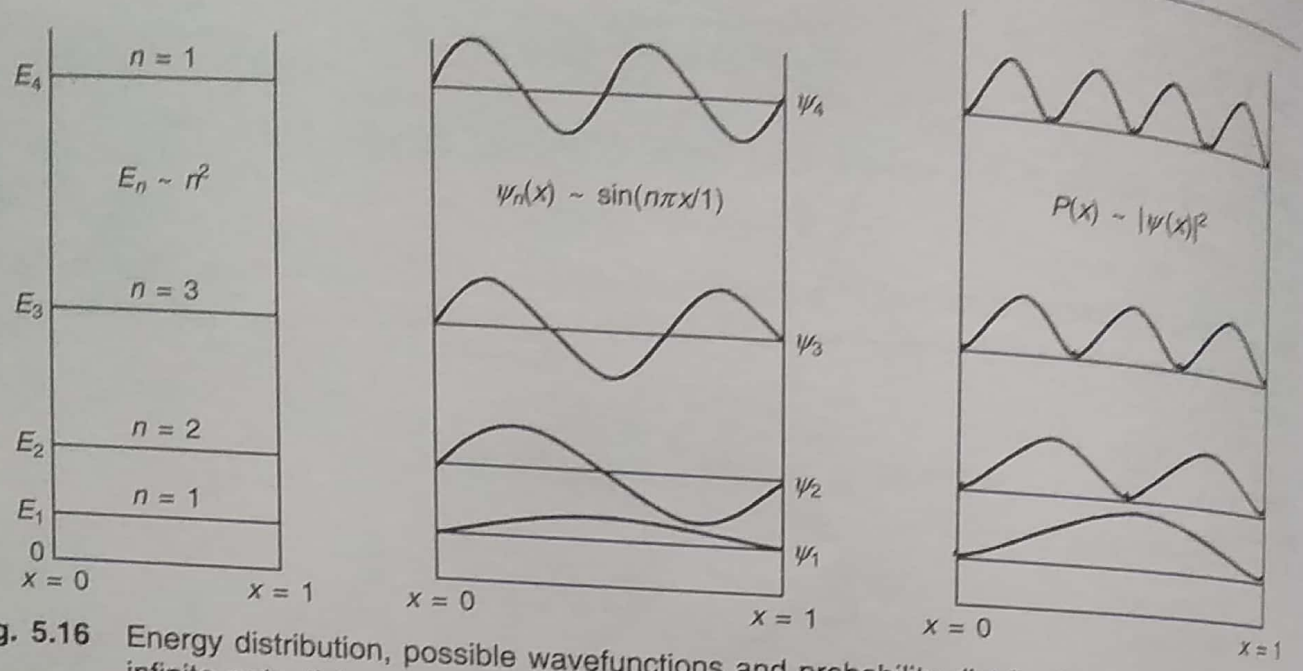


Fig. 5.16 Energy distribution, possible wavefunctions and probability distributions of electron in 1D infinite potential well.

Two non-interacting particles simultaneously present in the 1D potential well, can be described by a **two-particle wavefunction**. If the particles are present in two quantum states n_1 and n_2, then the two-particle wavefunction can be written as

$$\psi_{n_1 n_2}(x_1, x_2) = A^2 \sin\left(\frac{n_1 \pi x_1}{l}\right) \sin\left(\frac{n_2 \pi x_2}{l}\right) = \psi_{n_1}(x_1)\, \psi_{n_2}(x_2) \tag{5.52}$$

The corresponding probability density is

$$P_{n_1 n_2}(x_1, x_2) = \psi^*_{n_1 n_2}(x_1, x_2)\, \psi_{n_1 n_2}(x_1, x_2) \tag{5.53}$$

(b) Potential step: Reflection and tunnelling: Quantum leak

Let us consider a particle moving along positive x-axis towards a finite potential step (cf. Fig. 5.17) defined as

$V = 0$ for $x < 0$ [Region I]

and $V = V_0$ for $x \geq 0$. [Region II]

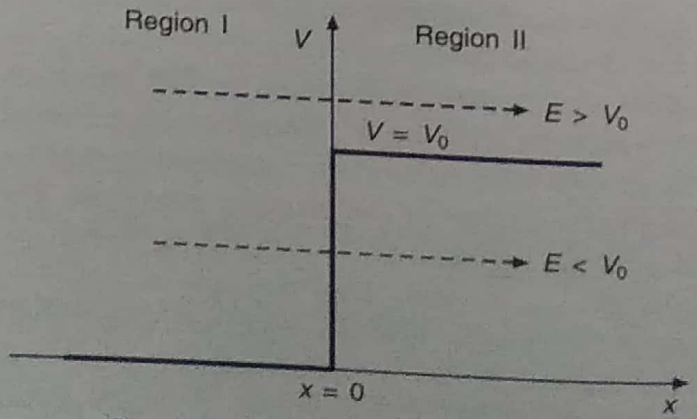


Fig. 5.17 Step potential at $x = 0$.

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Case I: $E > V_0$

If the incoming particle has energy greater than the potential step, then the Schrödinger equation will be

$$\frac{d^2\psi(x)}{dx^2} + k_1^2\,\psi(x) = 0 \qquad \text{[in Region I: } x < 0\text{]} \tag{5.54}$$

and

$$\frac{d^2\psi(x)}{dx^2} + k_2^2\,\psi(x) = 0 \qquad \text{[in Region II: } x \geq 0\text{]} \tag{5.55}$$

where $k_1^2 = 2mE/\hbar^2$ and $k_2^2 = 2m(E - V_0)/\hbar^2$; k_1, k_2 being real quantities.
The solutions of the above two equations are

$$\psi(x) = Ae^{ik_1x} + Be^{-ik_1x} \qquad \text{for } x < 0 \text{ [Region I]} \tag{5.56}$$

$$\psi(x) = Ce^{ik_2x} + De^{-ik_2x} \qquad \text{for } x \geq 0 \text{ [Region II]} \tag{5.57}$$

where A, B, C, D are arbitrary complex numbers. Physically, A denotes the amplitude of incident wave $\psi_{in}(x) = Ae^{ik_1x}$, moving along +ve X-direction. B is the amplitude of the reflected wave $\psi_{ref}(x) = Be^{-ik_1x}$ along –ve X-direction in Region I. C is the transmitted wave $\psi_{Tr}(x) = Ce^{ik_2x}$ in Region II moving along +ve X-axis. Since no particles are incident from right, $D = 0$. Also applying the boundary conditions of continuity for $\psi(x)$ and its derivatives at $x = 0$, we get, $A + B = C$ and $A - B = (k_1/k_2)C$, which give

$$\frac{B}{A} = \frac{k_1 - k_2}{k_1 + k_2} \tag{5.58}$$

and

$$\frac{C}{A} = \frac{2k_1}{k_1 + k_2} \tag{5.59}$$

The Transmission coefficient (T) and Reflection coefficient (R) are defined as the probability ratios of transmitted and reflected particle-waves with respect to that of incident particle-wave respectively. Therefore, one can get the expressions of T and R as

$$T = \frac{k_2}{k_1}\left|\frac{C}{A}\right|^2 = \frac{4k_1k_2}{(k_1 + k_2)^2} \tag{5.60}$$

and

$$R = \left|\frac{B}{A}\right|^2 = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2 \tag{5.61}$$

It is to be noted that $T + R = 1$, which means the total probability always equals to unity. These expressions are similar to the laws of reflection and transmission of light between the two media of varied refractive index. In this case of particle waves, the expressions physically mean that a ***quantum particle will be partially reflected even if it has enough energy to propagate past the step potential*** V_0. This condition typically illustrates the nanophysical behaviour of quantum world which is quite different from the classical conditions.

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Case II: $E < V_0$

When the incoming particle has energy less than the step potential, Schrödinger equation in Region I will be same but in Region II, it will look like

$$\frac{d^2\psi(x)}{dx^2} - k_3^2\,\psi(x) = 0 \tag{5.62}$$

where $k_3^2 = \dfrac{2m\,(V_0 - E)}{\hbar^2} = -\,k_2^2$; k_3 being a real decay constant. The solutions will be similar to the previous one as

$$\psi(x) = A_1 e^{ik_1 x} + B_1 e^{-ik_1 x} \quad \text{for x} < 0 \text{ [Region I]} \tag{5.63}$$

$$\psi(x) = C_1 e^{-k_3 x} + D_1 e^{k_3 x} \quad \text{for } x \geq 0 \text{ [Region II]} \tag{5.64}$$

A_1, B_1, C_1 and D_1 are arbitrary complex numbers. Now, as $x \to \infty$, $e^{k_3 x} \to \infty$. Therefore, D_1 should be zero to prevent the particles from unphysically collecting at large positive x. Also applying the boundary conditions for $\psi(x)$ and its derivatives at $x = 0$, we get and $A_1 + B_1 = C_1$ and $A_1 - B_1 = (ik_3/k_1)C_1$, which give

$$\frac{B_1}{A_1} = \left(\frac{k_1 - ik_3}{k_1 + ik_3}\right)^2 \tag{5.65}$$

and

$$\frac{C_1}{A_1} = \frac{2k_1}{k_1 + ik_3} \tag{5.66}$$

Therefore, the reflection coefficient will be

$$R = \left|\frac{B_1}{A_1}\right|^2 = \left(\frac{k_1 - ik_3}{k_1 + ik_3}\right)^2 = 1 \tag{5.67}$$

The solution in Region II ($x > 0$) is now an exponentially decaying function. This function is not automatically zero in the region of –ve energy. If we find the probability at $x = 0$ as

$$|C|^2 = |A|^2 \frac{4k_1^2}{(k_1^2 + k_3^2)^2} = |A|^2 \frac{4E}{E + (V_0 - E)} = |A|^2 \frac{4E}{V_0} \tag{5.68}$$

we see that this probability is finite (as $E < V_0$). Therefore, the probability of finding the particle in the forbidden region of positive x can be done by integrating the wavefunction over the x values from 0 to ∞, which will be finite and proportional to $|C|^2$. This is an example of ***nanophysical tunnelling effect, a particle can be sometimes found where its energy is negative ($E < V_0$) and, therefore, classically forbidden***. Actually, the wave penetrating a small distance from the boundary into the Region II is continually reflected till all the incident energy is turned back into Region I. Due to such continual reflection, the amplitude of the wave penetrating into Region II falls off exponentially. Classically, a particle of energy $E < V_0$ can never penetrate into

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Region II, since its KE would be negative, but in nanophysical world there is a finite probability of the particle wave penetrating a short distance into the classically forbidden Region II and decaying exponentially.

Now for the case of $E > V_0$, according to Eq. (5.59), $|C| > |A|$ since $k_1 > k_2$. So the amplitude of the transmitted wave is greater than that of incident wave. Also, as the KE of the particle is greater in Region I, therefore, the de Broglie wavelength is shorter in Region I with respect to Region II as explained in Fig. 5.18. On the other hand, for the case of $E < V_0$, the transmitted wave decays exponentially within a very short distance of Region II as explained earlier and shown in Fig. 5.18.

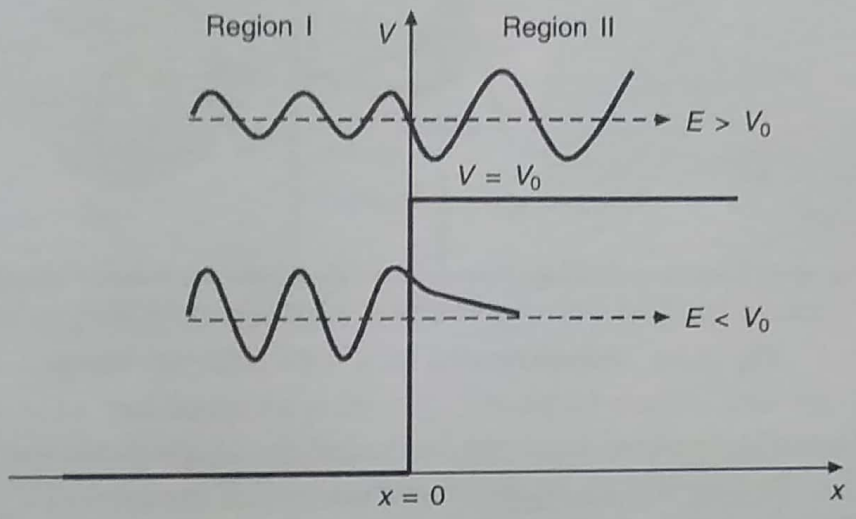


Fig. 5.18 Wavefunctions for a step potential.

Also the incident and reflected particle waves in the Region I ($x < 0$) add to form standing wave

$$\psi(x) = 2|A|\cos(kx + \varepsilon) \quad \text{(from Eq. (5.63), with } |B| = |A|) \tag{5.69}$$

where ε is the phase shift of the standing wave. This standing wave will give 'ripples' whenever a particle wave is reflected from a barrier, with a wavelength of $\lambda = 2\pi/k$ [where $k = (2mE/\hbar^2)^{1/2}$]. Such ripples are evident in Fig. 5.19.



Fig. 5.19 Scattered electrons formed standing wave 'ripples' inside a ring of Fe atoms on a Cu surface.

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(c) Specific case of tunnelling: Penetration of barrier

If the width of the step potential becomes finite (say, the potential step extends from $x = 0$ to $x = d$, i.e. the thickness of the potential barrier is d), then we will get the tunnelling phenomenon of barrier penetration (also called 'quantum leak'). Figure 5.20 describes finite potential barrier of width d and height V_0. A particle of energy $E < V_0$ in Region I when incident on the left side

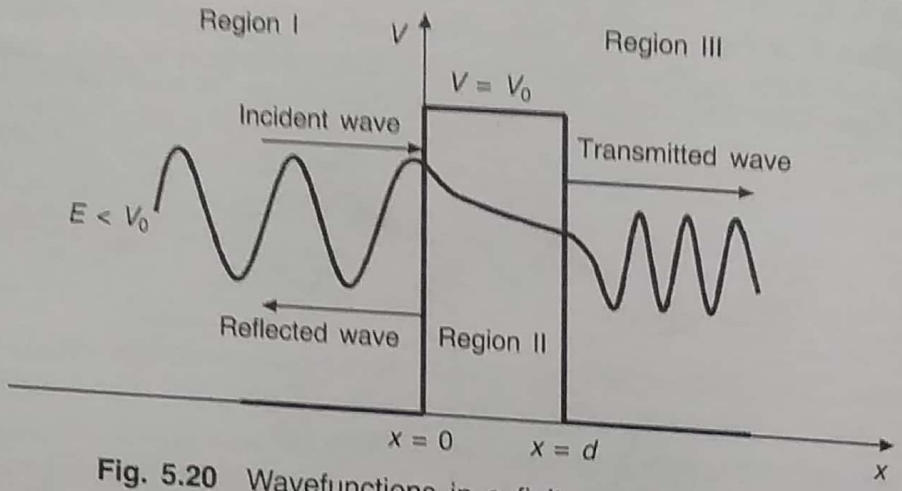


Fig. 5.20 Wavefunctions in a finite potential barrier.

of the barrier, a part of the particle wave gets reflected and is rest penetrates through the barrier and is transmitted to Region III. The tunnelling transmission probability T will be proportional to the ratio of the square of the amplitudes of transmitted wave ($\psi_{tr} = Fe^{ik_3x}$) and the incident wave ($\psi_{in} = Ae^{ik_1x}$). Therefore, we get

$$T = \frac{|F|^2}{|A|^2} \tag{5.70}$$

This problem can be solved in a similar way as described earlier with proper formation of Schrödinger equations and using boundary conditions for Regions I, II and III, which would yield

$$T = \left[1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2(k_3 d)\right]^{-1} \tag{5.71}$$

Classically, the particle cannot go to Region III penetrating through the Region II if its energy is less than the barrier height ($E < V_0$). But in nanophysics, the particle has a finite probability to tunnel through the barrier. As E is less than V_0, T will monotonically decrease due to the factor $E(V_0 - E)$ as shown in Fig. 5.20. For the case of $k_3D >> 1$, the expression for transmission coefficient will be reduced to

$$T = \frac{16E(V_0 - E)}{V_0^2} (e^{-2k_3 d}) \tag{5.72}$$

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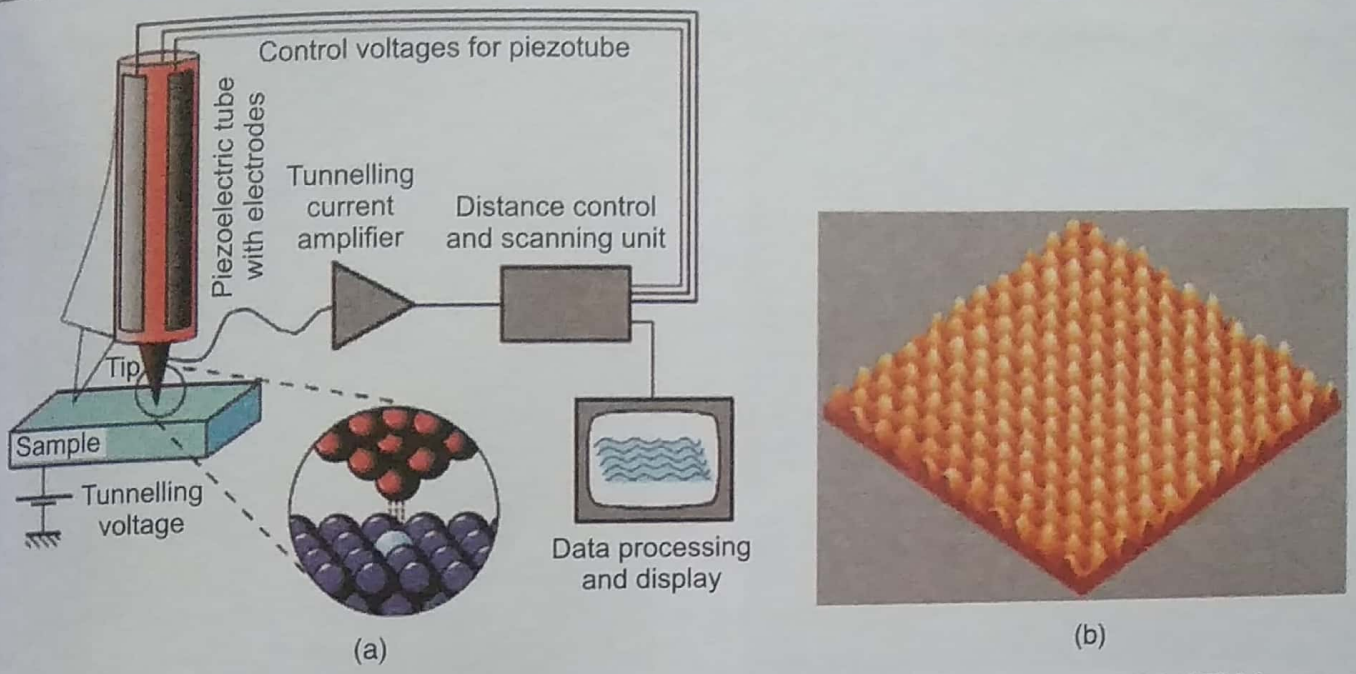


Fig. 5.21 **(a)** Schematic diagram of a Scanning Tunnelling Microscope [12], and **(b)** STM image of a graphite surface showing highly oriented pyrolytic graphite atoms.

where the symbols have the same meanings as described earlier and the exponential term provides the dominant dependence. Application of this nanophysical tunnelling is used in Scanning Tunnelling Microscope (STM), as shown in Figs. 5.21(a) and (b), to detect the surface profile of nanostructured materials. This instrument has given experimental access to the fascinating nano-scale world. This technique offers the opportunity to image conducting and semiconducting surfaces and to perform tunnelling spectroscopy with atomic-scale spatial resolution. It is a non-optical microscopy technique which employs principles of quantum mechanical barrier tunnelling, as explained above. A sharp probe (the tip), whose end is as sharp as a single atom, moves over the surface of the material under study, and a voltage is applied between the probe and the sample surface. Depending on the voltage applied, electrons will tunnel through the potential barrier between the surface and probe, resulting in a weak electric current. The direction of the tunnelling depends on the polarity of the electric field. The magnitude of this current is exponentially dependent on the distance between the probe and the surface (cf. Eq. (5.72)) and carry the information of the surface profile of the material under test.

(d) Potential box: Trapped particle in 3D: Nanodot

If we consider a particle inside a 3D infinite trap (Fig. 5.22) of volume L^3 with impenetrable walls, then we need to consider the generalized Schrödinger equation in three dimensions as

$$-\left(\frac{\hbar^2}{2m}\right)\nabla^2\psi(\vec{r})+V(\vec{r})\,\psi(\vec{r})=E\psi(\vec{r}) \tag{5.73}$$

where $\nabla^2=\dfrac{\partial^2}{\partial x^2}+\dfrac{\partial^2}{\partial y^2}+\dfrac{\partial^2}{\partial y^2}$ and $\vec{r}$ is the position vector. And the corresponding solutions will be

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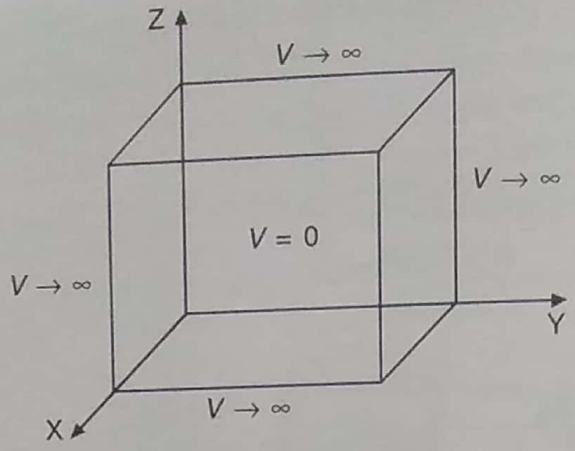


Fig. 5.22 Three-dimensional infinite potential well.

$$\psi_n(x,y,z) = \left(\frac{2}{L}\right)^{3/2} \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right) \tag{5.74}$$

and

$$E_n = \left[\frac{h^2}{8mL^2}\right](n_x^2 + n_y^2 + n_z^2) \tag{5.75}$$

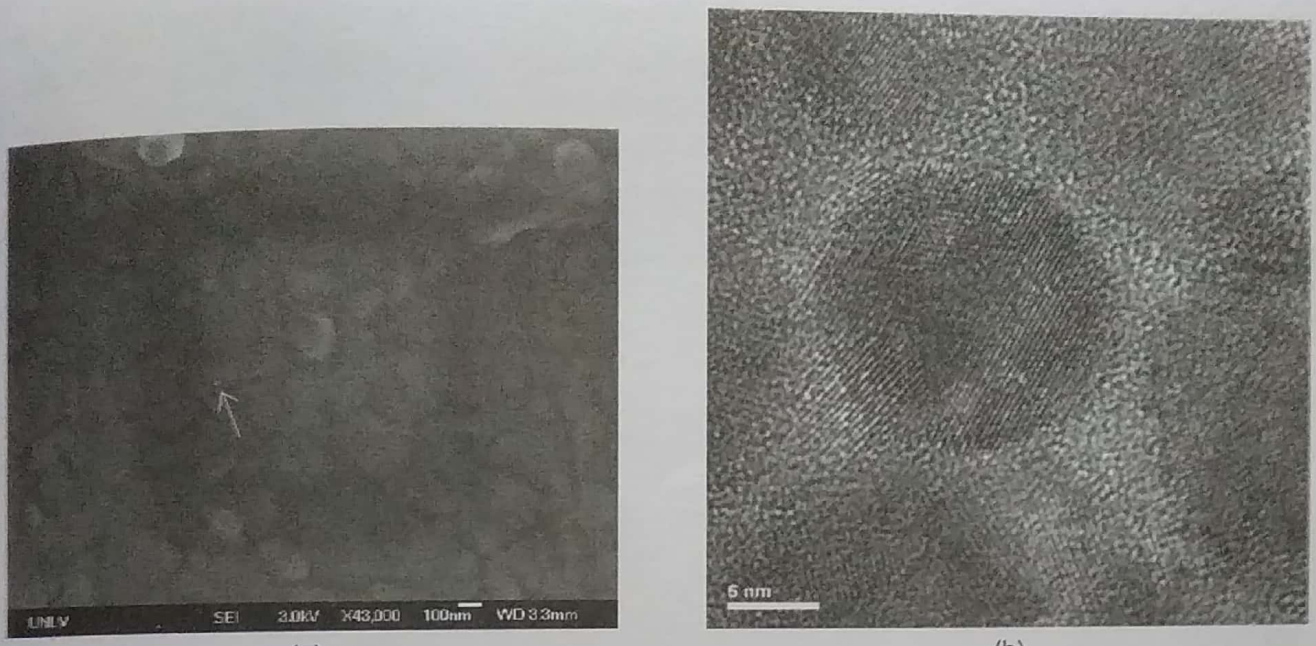
where n_x = 1, 2, 3,..., etc. These results can also be used for boxes of unequal dimensions L_x, L_y and L_z as

$$E_{n_x n_y n_z} = \left[\frac{h^2}{8m}\right]\left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}\right) \tag{5.76}$$

It is apparent from the above equation that the energy is dependent on three quantum numbers n_x, n_y and n_z. The lowest energy is E_{111} and non-zero (for n_x = 1, n_y = 1 and n_z = 1). The next energy level corresponds to E_{211} and is equivalent to E_{121} and E_{112} (for either one of the n-values as 2 and the other two as 1). Therefore, the second energy level has three states for same energy as ψ_{211}, ψ_{121} and ψ_{112}. The number of states that have the same energy is called degeneracy of that energy level. Hence the second energy level is thus three-fold degenerate.

Nanoparticles with a fraction of a nanometer to a few tens of nanometers size can be treated as the classic examples of three-dimensional confinement of carriers in a small 3D space. Figure 5.23(a) shows the Field Emission Scanning Electron Microscopy (FESEM) image of a Si nanoparticle (shown by the arrow). The particle diameter is around 15 nm. Figure 5.23(b) shows the High Resolution Transmission Electron Microscopy (HRTEM) image of a similar kind of Si nanoparticle of diameter around 15 nm. Highly oriented atomic planes are observed in different parts of the nanoparticle [13]. Carriers are confined within the 3D space of this nanoparticle and discrete energy levels will be formed whose expressions will be similar to Eq. (5.76), with a little modification due to quantum confinement effect, which will be discussed later.

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(a) (b)

Fig. 5.23 **(a)** FESEM image of a single Si nanoparticle shown by the arrow, and **(b)** HRTEM image of a 15 nm Si nanoparticle. Oriented atomic planes are visible inside the nanoparticle [13].

(e) Electron trapped in 2D plane: Nanosheet

The abovementioned results can also be used for 2D spaces (Fig. 5.24) where carriers are confined in one-dimension, say along Z-direction to a very small thickness t, and free to move along X- and Y-directions. In nanophysics, this is the case of nanosheets where carriers are allowed to move along a two-dimensional plane. In this case, the modification of wavefunction and energy will respectively be

$$\psi_n(x, y, z) = \left(\frac{2}{t}\right)^{1/2} \sin\left(\frac{n_z \pi z}{t}\right) e^{ik_x x} e^{ik_y y} \tag{5.77}$$

and

$$E_n = \left[\frac{h^2}{8mt^2}\right] n_z^2 + \frac{\hbar^2 k_x^2}{2m} + \frac{\hbar^2 k_y^2}{2m} \tag{5.78}$$

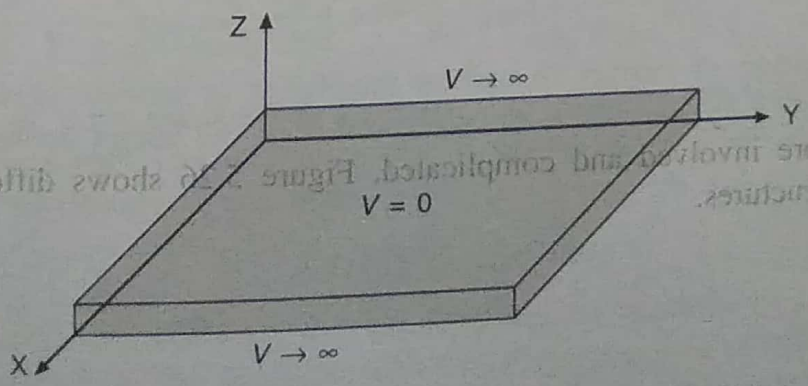


Fig. 5.24 Two-dimensional space. Particles can move along X- and Y-directions freely but confined along Z-direction by a small distance t.

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where the quantum number n_z is called the sub-band index and for $n = 1$ the carrier is assumed to be in the first sub-band. In general, the electron motion in a semiconductor band is described by introducing the effective mass of electrons (and holes) [14]. Figure 5.19 shows an STM picture of a 2D circular potential well. The well is generated by a circular array of Fe atoms on Cu surface and the free electrons on the Cu surface are trapped inside this 2D potential well.

(f) Electrons moving in one-dimension: Nanowire/rod/belt

As shown in Fig. 5.25, in nanowire, carriers are confined in both directions (say, Y and Z) to

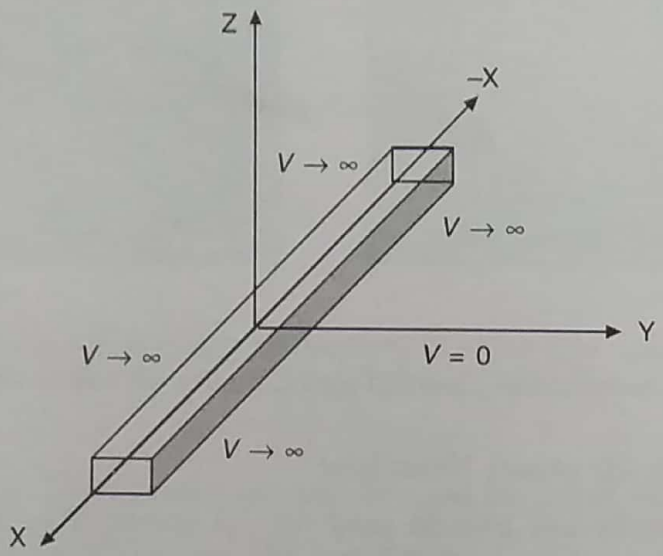


Fig. 5.25 One-dimensional nanowire. Particles are only allowed to move freely along X-axis and confined along Y- and Z-directions by a small distance t (assuming square cross-section).

small dimension, t (assuming square cross-section, t^2) and allowed to move freely along the length of x-axis. In this case, the wavefunction and the energy will respectively be

$$\psi_{nn}(x, y, z) = \left(\frac{2}{t}\right) \sin\left(\frac{n_y \pi y}{t}\right) \sin\left(\frac{n_z \pi z}{t}\right) e^{ik_x x} \tag{5.79}$$

and

$$E_n = \left(\frac{h^2}{8md^2}\right)(n_y^2 + n_z^2) + \frac{h^2 k_x^2}{2m} \tag{5.80}$$

In nanophysics, motion of carries in carbon nanotubes or similar kind of nanowires can qualitatively be treated by the above equation, although the actual bound states and energy expressions are more involved and complicated. Figure 5.26 shows different types of one-dimensional nanostructures.

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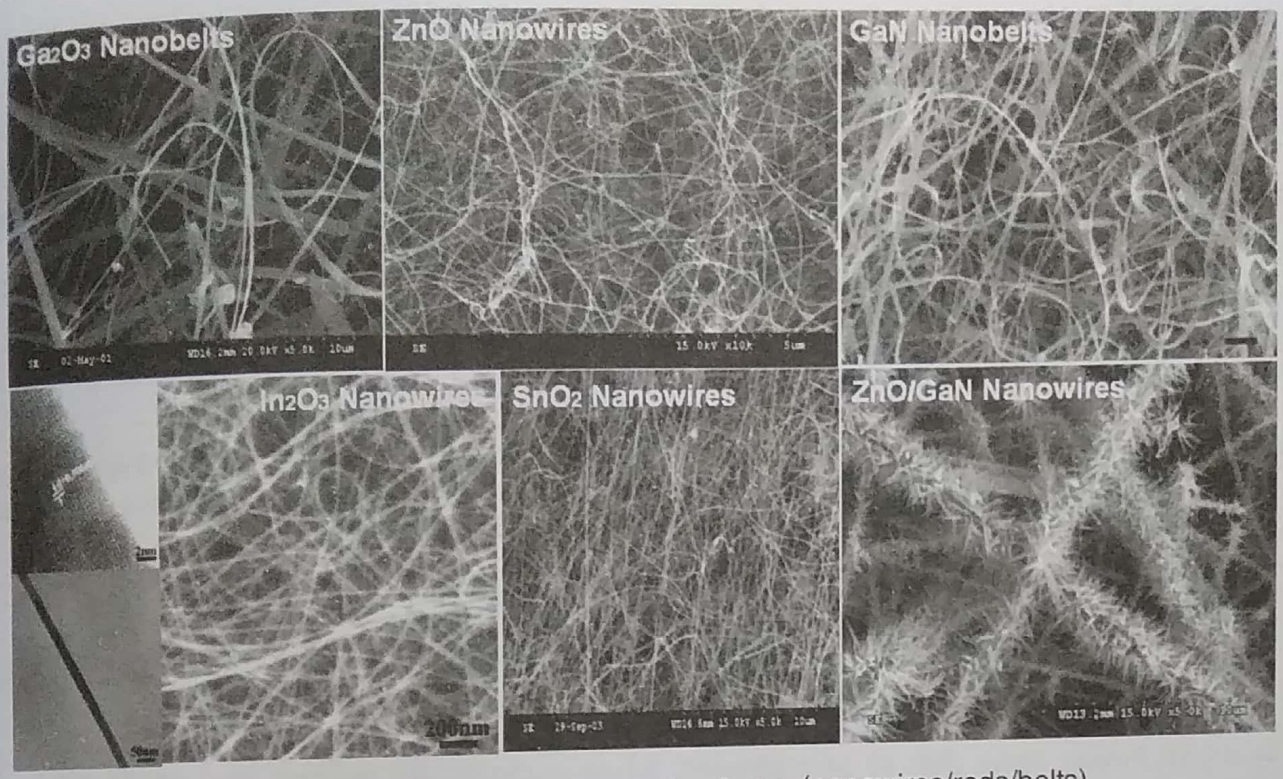


Fig. 5.26 One-dimensional nanostructures (nanowires/rods/belts).

(g) One-electron atoms: The hydrogen atom

The motion of electrons in the central field is one of the interesting problems in nanophysics. Though the field is not exactly Coulombian in character in most atoms because of the influence of other electrons, but in one-electron atoms (e.g. hydrogen-like atoms), the field may be regarded as Coulombian. The potential energy is given by

$$V(r) = -k\frac{Ze^2}{r} \tag{5.81}$$

where Ze is the nuclear charge, r is the position of the electron in spherical polar co-ordinate of central field and k is the Coulomb constant (cf. Fig. 5.27). The Schrödinger equation for this system (in spherical polar co-ordinate) will be

$$\frac{-\hbar^2}{2m}\frac{1}{r^2}\frac{\partial}{\partial r}\left(\frac{r^2\partial\psi}{\partial r}\right) - \frac{\hbar^2}{2mr^2}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}\right] + \left(-k\frac{Ze^2}{r}\right)\psi = E\psi \tag{5.82}$$

where $\psi \equiv \psi(r, \theta, \phi)$, $x = r \sin\theta\cos\phi$, $y = r \sin\theta \sin\phi$, $z = r\cos\theta$ and θ, ϕ, are polar and azimuthal angles. From spherical symmetry $\psi(r, \theta, \phi)$ can be separated into three independent functions of single variables r, θ and ϕ as

$$\psi(r,\theta,\phi) = R(r)\,\Theta(\theta)\,\Phi(\phi) \tag{5.83}$$

By solving the Schrödinger equation, the solutions can be described by the quantum states $\Psi_{n,l,m_l}(r, \theta, \phi)$, specified by the quantum numbers n (principal quantum number), l (orbital quantum number) and m_l (magnetic quantum number) and the acceptable solutions (i.e. Ψ_{n,l,m_l} is continuous, single-valued, with continuous gradient) will be obtained by assigning certain allowed values of n, l and m_l, furnished as follows.

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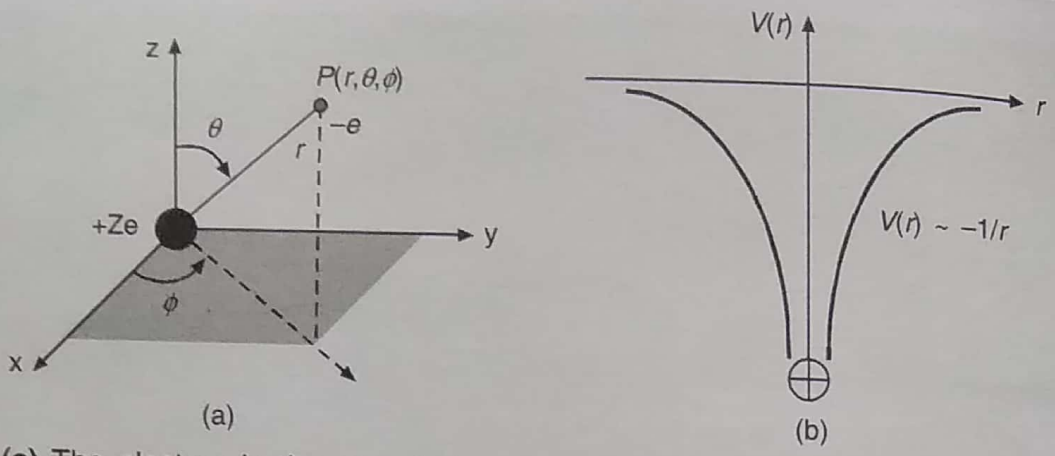


Fig. 5.27 **(a)** The electron in the one-electron atom is attracted by a central force that is always directed towards the positive nucleus and described by spherical coordinates r, θ, ϕ, and **(b)** variation of potential energy with respect to the nucleus.

$$\left.\begin{aligned}&\text{Principal quantum number: } n = 1, 2, 3,\ldots\\&\text{Orbital quantum number: } l = 0, 1, 2,\ldots, (n-1)\\&\text{Magnetic quantum number: } m_l = 0, \pm1, \pm2, \pm3,\ldots, \pm l\end{aligned}\right\} \quad (5.84)$$

It is worth noting again that in nanophysics, when particles are trapped in a small region of space, quantum numbers play significant role for the proper description of the system. To exhibit the dependence of $R(r)$, $\Theta(\theta)$, $\Phi(\phi)$ upon the quantum numbers n, l and m_l we may re-write the electron wavefunction of a hydrogen-like atom as

$$\Psi_{n,l,m_l}(r,\theta,\phi) = R_{n,l}(r)\,\Theta_{l,m_l}(\theta)\,\Phi_{m_l}(\phi) \quad (5.85)$$

The wavefunctions $R(r)$, $\Theta(\theta)$ and $\Phi(\phi)$ together with Ψ_{n,l,m_l} are given in Table 5.1 for $n = 1$ and 2. The parameter a_0 given in the table is identical to its value in Bohr model (= 0.053 nm, cf. Section 5.3.1) but no longer signifies the exact radius of an orbit. Also in literature $l = 0, 1, 2,\ldots$, etc. are denoted by s, p, d,... etc. Therefore, 2p wavefunction has $n = 2$, $l = 1$. Figure 5.28 shows the surface plots of three p-orbitals. The energies of the one-electron system will be same as Bohr model and can be expressed as

$$E_n = -\frac{Z^2 E_0}{n^2} \quad (5.86)$$

where $E_0 = 13.6$ eV and Z is the atomic number.

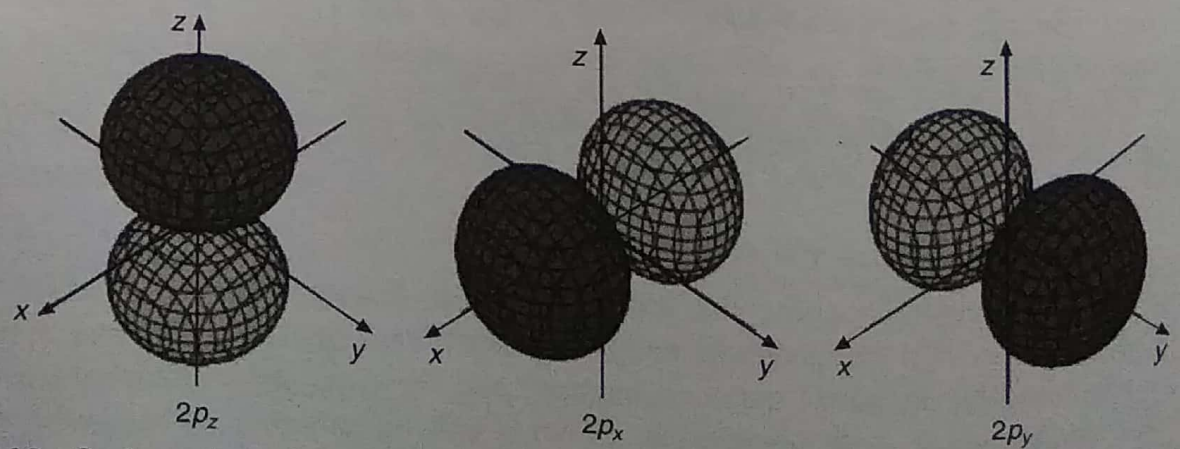


Fig. 5.28 Surface plots of the three 2p orbitals. The dark grey shading indicates a positive value for the wavefunction and light grey indicates a negative value.

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Table 5.1 Normalized wavefunction of hydrogen-like atoms for $n = 1$ and 2

n	l	m_l	$R_{n,l}(r)$	$\Theta_{l,m_l}(\theta)$	$\Phi_{m_l}(\phi)$	$\psi_{n,l,m_l}(r,\theta,\phi)$	Wave-function name, real form
1	0	0	$\frac{2}{a_0^{3/2}} e^{-r/a_0}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$	$\psi_{100} = \frac{1}{\sqrt{\pi} a_0^{3/2}} e^{-r/a_0}$	1s
	0	0	$\frac{1}{2\sqrt{2} a_0^{3/2}} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$	$\psi_{200} = \frac{1}{4\sqrt{2\pi} a_0^{3/2}} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$	2s
2	1	0	$\frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	$\frac{\sqrt{6}}{2} \cos\theta$	$\frac{1}{\sqrt{2\pi}}$	$\psi_{210} = \frac{1}{4\sqrt{2\pi} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0} \cos\theta$	$2p_z$
	1	±1	$\frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	$\frac{\sqrt{3}}{2} \sin\theta$	$\frac{1}{\sqrt{2\pi}} e^{\pm i\phi}$	$\psi_{21\sin\phi/\cos\phi} = \frac{1}{8\sqrt{\pi} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0} \sin\theta\, e^{\pm i\phi}$	$2p_y$, $2p_x$

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(h) Excitons

The excitons are bound states of photo-excited and coulomb-correlated electron–hole pairs, produced momentarily by light illumination of energy greater than the bandgap of the material concerned [15,16]. The electron–hole pairs, thus produced, will orbit around each other, having equal and opposite charges (cf. Fig. 5.29). These electron–hole pairs are elementary excitation, or a quasi-particle of a solid and combine in a short time giving off a flash of light. As the exciton results from the binding of the electron with its hole, the exciton has slightly less energy

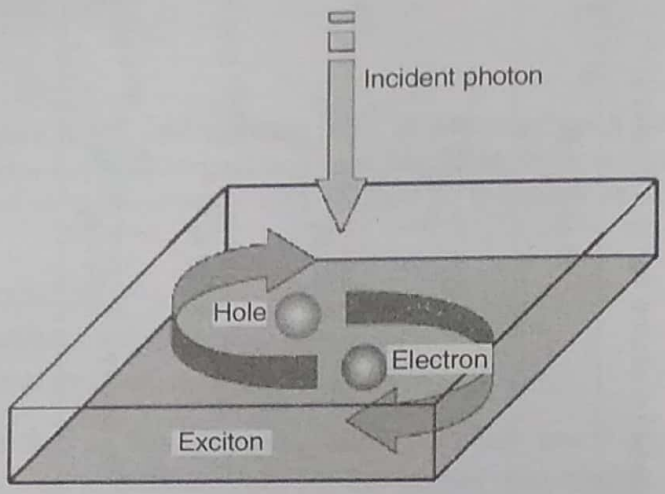


Fig. 5.29 Schematic diagram of the formation of an exciton inside a material.

than the unbound electron and hole. The wavefunction of the bound state is hydrogenic. However, the binding energy is much smaller and the size is much bigger than a hydrogen atom because of the effects of screening and the effective mass of the constituents in the material. Also unlike in the hydrogen atom, where the nucleus is assumed to be stationary and electrons circle around the nucleus, in the case of excitons, the electron–hole pairs circle around each other with respect to their centre of mass. Therfore, the Schrodinger equation should contain terms involving both electron and hole. Hence the equation will look like (in Cartesian coordinate).

$$-\left(\frac{\hbar^2}{2m_h^*}\nabla_h^2+\frac{\hbar^2}{2m_e^*}\nabla_e^2\right)\psi-k\frac{e^2}{|\vec{r}_e-\vec{r}_h|}\psi=E\psi \tag{5.87}$$

where m_h^* and m_e^* are the effective masses of the hole and the electron respectively, $\vec{r}_e$ and $\vec{r}_h$ are the position vectors for the electron and the hole with respect to the centre of mass of the system and k is the coulomb constant, which involves the dielectric constant of the material involved.

Depending on the materials in question, there are two types of excitons that can exist. In semiconductors, the dielectric constant is generally large, and as a result, screening tends to reduce the Coulomb interaction between electrons and holes. This results in the formation of **Mott–Wannier exciton**, which has a radius much larger than the lattice spacing. As a result, the effect of the lattice potential can be incorporated into the effective masses of the electron and the hole, and because of the lower masses and the screened Coulomb interaction, the binding energy is usually much less than a hydrogen atom, typically of the order 0.1 eV. On the other hand, when the dielectric constant of the material is very small, the Coulomb interaction between

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the electron and the hole becomes very strong and the excitons tend to be much smaller, of the same order as the unit cell, so the electron and the hole sit on the same cell. This type of excitons are called **Frenkel excitons**, whose binding energy is typically of the order of 1.0 eV.

(i) Quantum confinement effect in nanomaterials

In terms of hydrogen atom analogy, the band gap is the ionization limit of the hydrogenic electron–hole bound states. Absorption of photon at higher energies creates free electrons and holes with excess kinetic energies inside the semiconductor. Also the dielectric constants of most of the inorganic semiconductors range from 5 to 12, which implies that the Coulomb interaction is almost screened. Also, the small effective masses of electrons and the holes inside the semiconducting material imply that the localization energies for the electrons and the holes are large. Thus the excitation wavefunction extends over a large region within the material. For example, the lowest 1s exciton of CdS has an approximate diameter of 6 nm. Therefore, when the crystallite size is comparable to this dimension, 'bulk' energy level scheme will not be valid and electron–hole interactions will dominate the dynamics. So in the nanometric dimensions of the material, the energy level scheme depends upon the size, shape and nature of the material. Proper formation of the potential energy terms along with the Schrödinger equation and wavefunctions show that the bandgap of the nanocrystalline material increases with respect to the bulk value, i.e. there will be a blue-shift (ΔE) of the bandgap as the crystallite size approaches from 'bulk' size to lower dimensions of nanometric range as shown below:

$$\Delta E = E_{g(\text{nano})} - E_{g(\text{bulk})} = \frac{h^2 n^2}{8\mu^* r^2} - \frac{1.8e^2}{\varepsilon r} - 0.248E_R^* \tag{5.88}$$

where r is the radius of the nanoparticle, n is an integer, $\mu^* = [(m_h^*)^{-1} + (m_e^*)^{-1}]^{-1}$ is the reduced mass of the effective masses of the electron and the hole respectively, ε is the dielectric constant of the semiconducting material, E_R^* is the effective Rydberg energy. The first term of the right-hand-side expression in the equation represents the particle-in-a-box quantum localization energy and has a $1/r^2$ dependence for both the electron and the hole. The second term represents the Coulomb energy with an $1/r$ dependence and the third term is responsible for the spatial correlation. The n values larger than 1 in Eq. (5.88) correspond to the exciton transitions to higher excited states. In the limit of large r, the value of $E_{g(\text{nano})}$ approaches that of $E_{g(\text{bulk})}$. This bandgap enhancement in nanomaterials is experimentally observed in many semiconducting materials as shown in Fig. 5.30 and described in Table 5.2.

Table 5.2 Increase in the optical band-gap of p-$CuAlO_2$ nanoparticles with decrease in the average particle size [17], showing the quantum confinement effect of the band-gap enhancement in semiconductor nanocrystals

Particle size (nm)	*Band gap* (eV)
> 90 (bulk)	3.34
60	3.60
30	3.72
20	3.84
10	3.94

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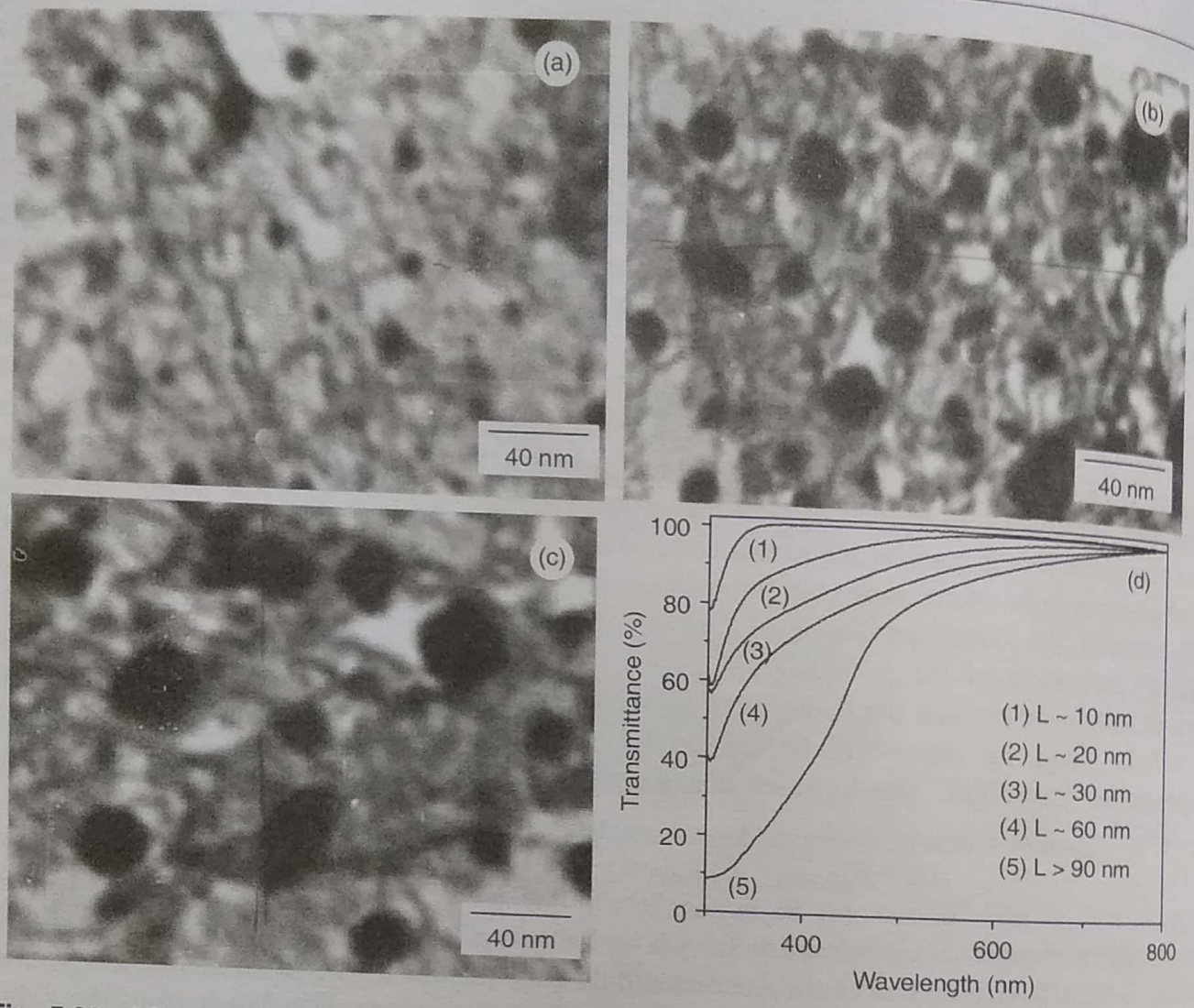


Fig. 5.30 TEM images of p-$CuAlO_2$ nanoparticles of average size (L) of **(a)** 10 nm, **(b)** 20 nm and **(c)** 30 nm, and **(d)** optical transmittance spectra of the nanoparticles showing the blue-shift of the band-gap edges with decrease in the particle size, indicating the size-effects of band-gap enhancement in semiconductor nanocrystals.

The optical and luminescent properties of nanoparticles are greatly dependent on this quantum size effect. As shown in Fig. 5.6, the CdSe quantum dots are found to emit different colours as their sizes change. As the size decreases, the emitted light wavelength moves towards the blue region. Thus, by changing the size of the quantum dots, one can generate different wavelengths of light through them under irradiation. Similarly, Fig. 5.31 shows the size effect in the photoluminescence spectra of p-$CuAlO_2$ nanoparticles. As the size decreases, the emission peak shifts towards low wavelength (blue-shift) region.

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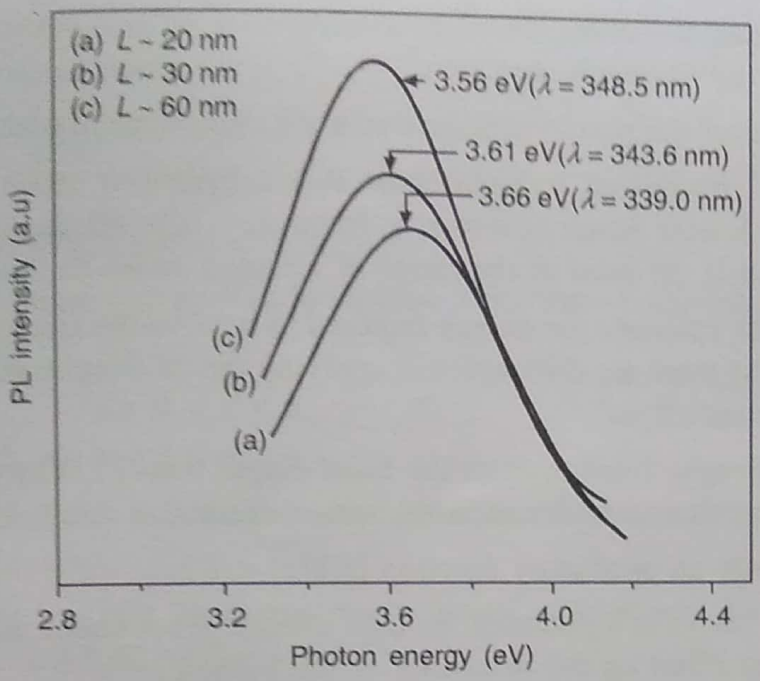


Fig. 5.31 Photoluminescence spectra of p-$CuAlO_2$ nanoparticles showing a blue-shift in the emission peak with a decrease in the particle size [17].

EXERCISES

1. Why is the wave nature of matter is not apparent to us in our daily observations?
2. The uncertainty principle is sometimes stated in terms of angular quantities as $\Delta L_\varphi \, \Delta\varphi \geq \hbar/2$, where ΔL_φ is the uncertainty in a component of angular momentum and $\Delta\varphi$ is the uncertainty in the corresponding angular position. In some quantum mechanical systems, the angular momentum is measured to have a definite (quantized) magnitude. Does this statement contradict the uncertainty principle?
3. Show that the de Broglie wavelength of a particle of charge e and rest mass m_0, moving at relativistic speed is given as a function of the accelerating potential V as $\lambda = (h/\sqrt{2m_0eV})[1 + (eV/2m_0c^2)^{-1/2}]$. Also show how this agrees with $\lambda = h/p$ in the nonrelativistic limit.
4. Show that for a free particle, the uncertainty relation can also be written as $\Delta\lambda \, \Delta x \geq \lambda^2/4\pi$.
5. Show that the smallest possible uncertainty in the position of an electron whose speed is given by $\beta = v/c$ is $\Delta x_{\min} = h/4\pi m_0 c \, (1 - \beta^2)^{1/2}$.
6. A microscope using photons is employed to locate an electron in an atom to within a distance of 0.2 Å. What is the uncertainty in the velocity of the electron located in this way?
7. A boy on the top of a table of h is dropping marbles of mass m to the floor and trying to hit a specific spot on the floor by using highest precision instrument. Show that the marbles will miss the spot by an average distance of the order of $(h/m)^{1/2}(h/g)^{1/2}$.

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8. Compare the gravitational attraction of an electron and proton in the ground state of a hydrogen atom to the Coulomb attraction.
9. Calculate the speed of the proton in a ground state of hydrogen atom.
10. Using the Balmer generalized formula show that a hydrogen series identified by the integer m of the lowest level occupies a frequency interval range given by $\Delta\nu = cR_H/(m+1)^2$. What is the ratio of the range of Lymann series to that of Pfund series?
11. Using Bohr's model, calculate the energy required to remove the electron from the singly ionized helium. Why there are difficulties in applying the de Broglie relation to a particle of variable linear momentum?
12. How does the de Broglie relation enter the Schrödinger theory? Why is it important for the Schrödinger equation to be linear in the wavefunction?
13. Why is ψ necessarily an oscillatory function if $V(x) < E$?
14. If a particle is not bound in a potential, its total energy is not quantized. Does this mean the potential has no effect on the behaviour of the particle?
15. If the wave function $\psi_1(x, t)$, $\psi_2(x, t)$ and $\psi_3(x, t)$ are three solutions to the Schrödinger equation for a particular potential $V(x, t)$, show that the arbitrary linear combination $\psi(x, t) = A_1\psi_1(x, t) + A_2\psi_1(x, t) + A_3\psi_1(x, t)$ is also a solution to that equation.
16. Consider a particle moving in the potential $V(x)$ shown in Fig. E5.1. For the following ranges of the total energy E, state whether there are any allowed values of E and if so, whether they are discretely separated or continuously distributed. (i) $E < V_0$, (ii) $V_0 < E < V_1$, (iii) $V_1 < E < V_2$, (iv) $V_2 < E < V_3$, (v) $V_3 < E$.

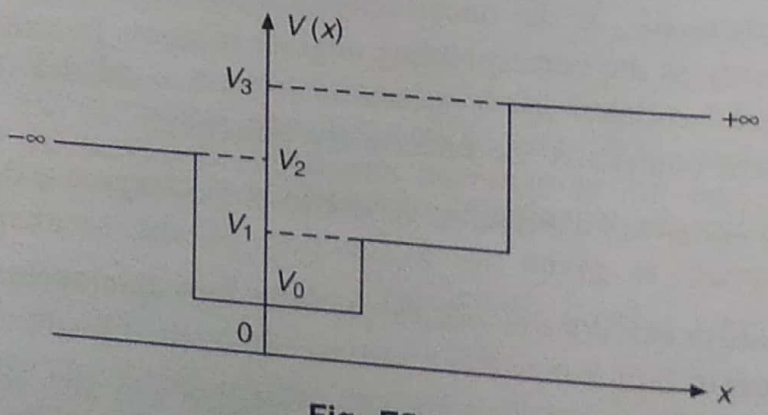


Fig. E5.1

17. An infinite potential well has a width of 10 nm. Inside the potential well, an electron has an effective mass $m^* = 0.067m_0$, where m_0 is the free electron mass. Calculate the lowest three allowable energy levels.
18. A cubic quantum dot (QD) of GaAs has a = 7.0 nm. Calculate the lowest three energy levels. Calculate the same if the dimensions of the QD are $a = 10$ nm, $b = 8$ nm and $c = 12$ nm and $m^* = 0.067m_0$.

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19. Estimate the lowest energy level for a one-dimensional infinite square well of width a containing cosine bump, i.e.

$$V = V_0 \cos(\pi x/a) \quad \text{for } -a/2 < x < +a/2$$

$$V \to \infty \quad \text{for } x < -a/2 \text{ or } x > +a/2$$

where $V_0 << \pi^2\hbar^2/2ma^2$.

20. A particle of total energy $9V_0$ is incident from the $-X$ axis on a potential given by

$$V = 8V_0 \quad \text{for } x < 0$$
$$= 0 \quad \text{for } 0 < x < a$$
$$= 5V_0 \quad \text{for } x > a.$$

Find the probability that the particle will be transmitted on through to the positive side of the X-axis, $x > a$.

21. For a particle in a box, show that the fractional difference in the energy between adjacent eigenvalues is $\Delta E_n/E_n = (2n + 1)/n^2$.

22. What is the relation between the size of a Bohr atom and the size of a Schrödinger atom?

23. Verify that the eigenfunction ψ_{211} and the associated eigenvalue E_2 satisfy the time-independent Schrödinger equation for the one-electron atom with $Z = 1$.

24. What are excitons? Why are excitonic effects more important in nanostructures than in bulk materials?

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[2] The Atomic Force Microscope (AFM) is one type of scanning probe microscopes, which is used to image surface structures on a nm or even sub-nm level and to measure surface forces. The standard AFM contains a microscopic tip (curvature radius of ~10—50 nm) attached to a cantilever spring. The underlying principle of AFM is the detection of the bending of this cantilever spring as a response to external forces. In the case of adhesive interaction between the tip and a surface, these forces are of the order 0.1—1 nN. To measure such small forces one must use not only very sensitive force-measuring springs but also very sensitive ways for measuring their bending. In order to detect this bending, which is as small as 0.01 nm, a laser beam is focused on the back of the cantilever. From there the laser beam is reflected towards a position-sensitive photodetector. Depending on the cantilever deflection the position of the reflected beam changes. The photodetector converts this change into an electrical signal (cf. Fig. 5.2(b)).

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[14] In solid state physics, a particle's **effective mass** is the mass it seems to carry in the semiclassical model of transport in a crystal. Under most conditions, electrons and holes in a crystal respond to electric and magnetic fields almost as if they were free particles in a vacuum, but with a different mass. This mass is usually stated in units of the ordinary mass of an electron m_e (9.11×10^{-31} kg). Quantum mechanical calculations give the expression of effective mass as $m^* = \hbar^2(d^2E(k)/dk^2)^{-1}$, where $E(k)$ is the energy as a function of k (wave number) and also called the dispersion relation. Effective masses of some common semiconductors are furnished below:

	Material	*Electron effective mass* (m_e^*)	*Hole effective mass* (m_h^*)
Group IV semiconductors	Si (4.2 K)	$1.08m_e$	$0.56m_e$
	Ge	$0.55m_e$	$0.37m_e$
III-V semiconductors	GaAs	$0.067m_e$	$0.45m_e$
	InSb	$0.013m_e$	$0.6m_e$
Group II-VI semiconductors	ZnO	$0.19m_e$	$1.21m_e$
	ZnSe	$0.17m_e$	$1.44m_e$

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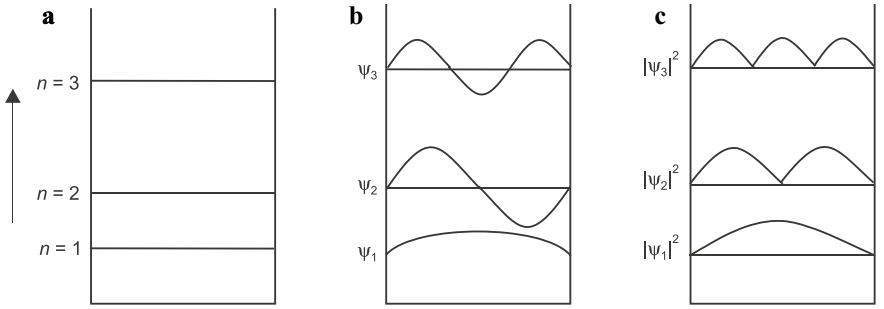


Fig. 1.8: (a) Quantized energy states, (b) corresponding wavefunctions and (c) probability of finding the particle at different locations between '0' and 'a' in the box.

Therefore

$$E_n = \frac{n^2\hbar^2\pi^2}{2ma^2} = \frac{n^2h^2}{8ma^2} \tag{1.39}$$

i.e.,

$$E_n \propto n^2 \tag{1.40}$$

Putting E_n in equation (1.37) we get

$$\Psi_n = A\sin\left(\frac{n\pi}{a}\right)x \tag{1.41}$$

Although n can take any integer value according to equation (1.39), in practice $n = 0$ and $\psi_n = 0$ are not allowed inside the box, because, if allowed, $|\psi|^2$ would be 0 and probability of finding the particle inside the box would be zero. For the same reason ψ cannot be zero inside the box. Therefore n takes the values $n = 1, 2, 3, \ldots$. This shows that energies of particle in a one dimensional potential box are quantized and can be illustrated as in Fig. 1.8(a).

Corresponding wave functions and probabilities of different states of particle in the box would look like those in Fig. 1.8(b) and (c) respectively.

1.5.2 Density of States

Density of states $D(E)$ is an important quantity. It enables to gain understanding of various spectroscopic and transport properties of materials. It is defined as the number of states per unit energy range and in general can be obtained as follows. Consider energy as is given for a particle in a 1D box.

$$E_n = n^2 \frac{h^2}{8ma^2} \quad \text{(Henceforth we drop suffix of } E\text{)}$$

$$dE = \frac{h^2}{8ma^2} \cdot 2n \cdot dn \tag{1.42}$$

$$\frac{dn}{dE} = \frac{8ma^2}{h^2} \cdot \frac{1}{2n} = \frac{a}{h}\sqrt{\frac{2m}{E}} \tag{1.43}$$

$$D(E) = \frac{dn}{dE} \propto E^{-1/2} \tag{1.44}$$

The equation (1.44) shows how the density of states will vary with change of energy. Depending upon the energy state's function, density of states would take form as briefly stated below.

Density of States for a Zero Dimensional (0-D) Solid

Neglecting the periodic potential existing in solids, we can imagine a zero dimensional solid in which electron is confined in a three dimensional potential box with extremely small (<100 nm) length, breadth and height as a 0-D solid. This will have the discrete energy levels as discussed above with density of states given as

$$D(E) = \frac{dN}{dE} = \sum_{\varepsilon_i} \delta(E - \varepsilon_i) \tag{1.45}$$

where ε_i are discrete energy levels and δ is Dirac function. The density of states as a function of energy would appear as illustrated in Fig. 1.9.

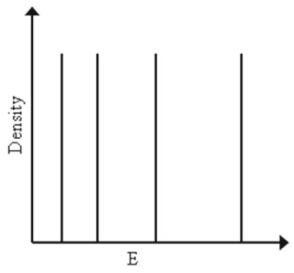


Fig. 1.9: Density of states for a particle in a zero dimensional solid.

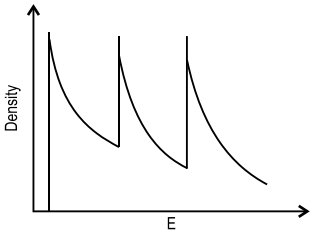


Fig. 1.10: Density of states for a particle in a one dimensional solid.

Density of States in a One Dimensional (1-D) Potential Box, Wire

A particle confined in one dimension is like a particle in a one dimensional potential well as in the previous case. However the potential in two directions is infinitely large but length is not very small. This gives rise to density of states as follows:

$$D(E) = \frac{dN}{dE} = \sum_{\varepsilon_i < E} \delta(E - \varepsilon_i)^{-1/2} \tag{1.46}$$

where ε_i are discrete energy levels. Figure 1.10 graphically illustrates nature of density of states for a one dimensional solid.

Density of States in a Two Dimensional (2-D) Potential Box, Thin Film

It can be shown that the density of states ($D(E)$) in the two dimensional solid, which is nothing but the case of thin films, is given as

$$D(E) = \frac{dN}{dE} = \sum_{\varepsilon_i < E} 1 \tag{1.47}$$

Thus it can be shown that the density of states in two dimensional case is constant (see Fig. 1.11). In this case $\frac{dN}{dE}$ would correspond to the states available in an area.

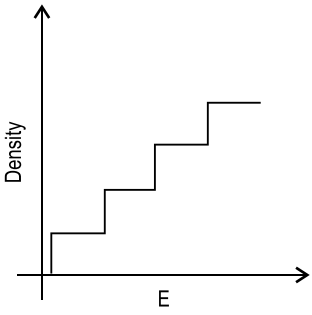


Fig. 1.11: Density of states for a 2D potential box.

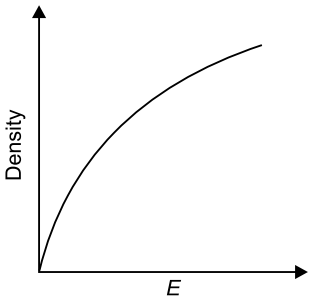


Fig. 1.12: Density of states for a particle in a 3-D potential box.

Density of States for a Particle in a Three Dimensional Box

Following the previous cases it is a straightforward task to show that in a box of finite length 'a', width 'b', and height 'c' with potential $V = 0$ inside the box and $V = \infty$ outside the box, the energy states can be obtained as

$$E_{n_x,n_y,n_z} = \frac{\hbar^2}{2m}\left(n_x^2 + n_y^2 + n_z^2\right) \tag{1.48}$$

The wave function has the form

$$\psi_n(x, y, z) = A \sin\left(\frac{\pi n_x x}{a}\right) \sin\left(\frac{\pi n_y y}{b}\right) \sin\left(\frac{\pi n_z z}{c}\right) \tag{1.49}$$

Hence to consider the density of states we shall require the finite volume. It can be shown that

$$D(E) \propto E^{1/2} \tag{1.50}$$

This is graphically illustrated in Fig. 1.12.

1.5.3 Particle in a Coulomb Potential

So far we have considered the particle confined to a potential box in which $V = 0$ inside the box and abruptly outside the box $V = \infty$. We shall consider now another type of potential viz. Coulomb potential experienced by a particle. The potential can be written as

$$V(r) = \frac{-Ze^2}{r} \tag{1.51}$$

This type of potential can arise when a particle moving in three dimensions is attracted towards the centre. For example an electron with negative charge attracted towards a positively charged nucleus at the centre of an atom.

In equation (1.51) 'Ze' is the charge due to Z number of electrons and r is the distance between nucleus and the charge. The negative sign arises by convention to show attraction between oppositely charged particles.

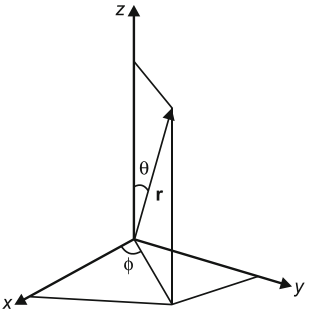
It is convenient to solve this problem in spherical coordinates rather than in Cartesian coordinates by writing ψ, the wave function in spherical coordinates $\psi(r, \theta, \varphi)$.

$$x = r \sin\theta \cos\phi \tag{1.52}$$

$$y = r \sin\phi \sin\theta \tag{1.53}$$

$$z = r \cos\ \theta \text{ and } r = \sqrt{x^2 + y^2 + z^2} \tag{1.54}$$

The time independent Schrödinger equation in spherical coordinates can be written as



$$\nabla^2\psi(\mathbf{r}, \theta, \phi) + \frac{2m}{\hbar^2}[E - V(\mathbf{r})]\psi(\mathbf{r}, \theta, \phi) = 0 \tag{1.55}$$

We shall not go here in the details of solving this equation but state the major steps and result. It is the spherically symmetric form of the potential which enables us to simplify the procedure.

Wavefunction $\Psi(\mathbf{r},\theta,\phi)$ is separable into functions of $\mathbf{r}$, θ, and φ as

$$\psi = R(r)Y(\theta, \varphi) \tag{1.56}$$

It can be written as

$$\psi_{n\ell m}(\mathbf{r}, \theta, \phi) = \frac{U_{n\ell}(r)}{r} Y_{\ell m}(\theta, \varphi) \tag{1.57}$$